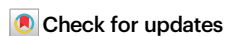


A structure-informed deep learning framework for modeling TCR-peptide-HLA interactions

Received: 18 March 2026

Accepted: 14 September 2026

Published online: 25 September 2026



Kai Cao¹, Rui Li^{1,2}, Martin Stražar¹, Eric M. Brown¹, Phuong N. U. Nguyen¹, Marie-Madlen Pust^{1,2}, Jihye Park¹, Daniel B. Graham^{1,2}, Orr Ashenberg^{1,3}, Caroline Uhler^{1,4}✉ & Ramnik J. Xavier^{1,2,3,5}✉

Interactions between T cell receptors (TCRs), peptides, and human leukocyte antigens (HLAs) underlie antigen-specific T cell immunity. Despite substantial advances in prediction methods, accurate modeling of coupled TCR–peptide–HLA recognition remains underdeveloped, limiting applications such as TCR and neoepitope prioritization in cancer and antigen identification in autoimmunity. Here we present StriMap, a unified framework for predicting TCR–peptide–HLA interactions by integrating physicochemical, sequence-context, and structural features at recognition interfaces. StriMap achieves state-of-the-art performance with improved generalizability and enables applications in cancer and autoimmunity. As a case study, we screened 13 million peptides from 43,241 bacterial proteins and identified candidate molecular mimics that were experimentally validated to activate T cells expressing an ankylosing spondylitis (AS)-associated TCR. A top validated peptide was enriched in patients with inflammatory bowel disease (IBD), suggesting potential shared microbial triggers. Overall, StriMap provides a framework for rational immunotherapy design and dissecting antigenic drivers of autoimmunity.

T cell-mediated immunity relies on the recognition of antigenic peptides presented by human leukocyte antigen (HLA) molecules through T cell receptors (TCRs) expressed on the surface of T cells¹. This HLA-restricted recognition of peptide epitopes is central to adaptive immunity, enabling the detection of infected or transformed cells while maintaining tolerance to self. This same recognition system, which enables broad immune surveillance, can also determine disease trajectories when dysregulated or redirected. Cancer and autoimmunity represent two challenging and important problems in which antigen-specific T cell recognition is central to both pathogenesis and therapy. In cancer, the ability to prioritize TCRs and neoepitopes underpins personalized immunotherapies and vaccines^{2,3}. In

autoimmunity, identifying disease-associated antigens and potential molecular mimics can reveal causal triggers and enable antigen-specific interventions^{4,5}.

While several experimental platforms can detect interactions between TCRs and peptide–HLA (pHLA) complexes^{6–8}, they remain time-consuming and expensive, limiting coverage of TCR or peptide diversity. These limitations motivate the need for computational approaches⁹. Computational models have advanced different stages of this recognition cascade. Peptide–HLA presentation predictors, such as NetMHCpan¹⁰, have become widely used for modeling antigen presentation^{11–14}, but they do not directly predict downstream TCR recognition. Systematically mapping downstream TCR–pHLA

¹Broad Institute of MIT and Harvard, Cambridge, MA, USA. ²Department of Molecular Biology, Massachusetts General Hospital and Harvard Medical School, Boston, MA, USA. ³Klarman Cell Observatory, Broad Institute of MIT and Harvard, Cambridge, MA, USA. ⁴Massachusetts Institute of Technology, Cambridge, MA, USA. ⁵Center for the Study of Inflammatory Bowel Disease, Massachusetts General Hospital, Boston, MA, USA.

✉ e-mail: cuhler@mit.edu; rxavier@broadinstitute.org

recognition remains underdeveloped⁹. This difficulty reflects the combinatorial complexity of the interaction space, the structural plasticity of recognition, and the limited and biased nature of available experimental data, as reflected in the most recent benchmark studies^{15–17}.

Deep learning has begun to be used to model these processes, yet many approaches rely primarily on sequence-based representations and treat pHLA presentation and TCR–pHLA recognition as independent tasks^{18–25}, potentially limiting generalizability beyond the training distribution by failing to capture their conditional dependence and shared structural constraints. More recently, structure-informed approaches, including SageTCR²⁶, NetTCR-struct²⁷, TRAP²⁸, STAG²⁹, and STAG-LLM³⁰, have demonstrated the value of incorporating structural information for modeling immune-recognition interfaces. However, structure-informed approaches often rely on a limited number of experimentally solved TCR–pHLA complex structures or on predicted full tri-molecular complex structures, whose accuracy and throughput remain challenging at flexible TCR–pHLA interfaces. These constraints can limit proteome-scale candidate screening and prospective experimental validation of newly predicted TCR–pHLA interactions. Thus, a scalable framework is needed that can incorporate structure-informed constraints from individual molecular chains without requiring accurate full-complex structure prediction, while explicitly linking peptide–HLA presentation with downstream TCR recognition and enabling experimentally testable candidate prioritization.

To address this gap, in this work, we develop StriMap (Structure-informed TRi-molecular Interaction Mapping), a unified deep learning framework that models antigen presentation and TCR recognition as a coupled biological process, integrating complementary sequence, structural, and contextual features to capture the determinants of immune recognition. Across diverse public datasets curated from multiple independent studies, StriMap consistently achieves state-of-the-art performance when benchmarked against leading HLA presentation and TCR-specificity models. We further demonstrate the utility of StriMap in both cancer and autoimmunity. In cancer, StriMap enables TCR-centric prioritization of recurrent neoepitopes and supports neoepitope selection for personalized vaccine design. In autoimmunity, we apply StriMap to investigate the link between HLA-B27-bound bacterial peptides and TCRs containing the conserved TRBV9 motif associated with ankylosing spondylitis (AS). Leveraging a small amount of recently published disease-specific data together with large-scale public TCR–pHLA datasets, we train the model to screen 13 million peptides derived from 43,241 bacterial proteins in gut bacterial strains linked to AS, ranking candidates by predicted binding scores. Notably, top-ranked peptides are experimentally validated to activate T cells expressing a representative TRBV9 TCR. Matched metagenomic and metatranscriptomic analyses reveal that one validated peptide from *Streptococcus* strains is enriched in IBD patients, suggesting a shared antigenic trigger between the two autoimmune diseases. Together, these results highlight StriMap as a generalizable framework for linking antigen presentation to TCR recognition across cancer and autoimmune disease, accessible via a web portal at (www.strimap.com).

Results

StriMap overview

To systematically model the tri-molecular interaction underlying T cell-mediated immunity (Fig. 1a), we developed StriMap, a unified deep learning framework that integrates physicochemical, sequence-context, and structural features. At the core of StriMap is the Sequence and Structure Feature Extractor (SSFE), a multimodal encoder that learns biophysical determinants of molecular recognition (Fig. 1b). For each input (peptide, HLA, and paired TCR α/β chains), the SSFE generates a comprehensive representation by fusing three

complementary channels: (i) residue-level physicochemical features³¹ (Supplementary Table 1); (ii) evolutionary sequence context from pretrained protein language models (e.g., ESM2³², ProtT5³³); and (iii) 3D structural features predicted by ESMFold³² and processed via an equivariant graph neural network (EGNN)³⁴ to encode spatial geometries (see “Methods”). A gated cross-attention is used to integrate these complementary multimodal representations at the residue level. We further introduce a bilinear attention network³⁵ (BAN) to model residue–residue interactions between molecular components (Fig. 1c). To reflect the hierarchical nature of immune recognition, StriMap employs a coupled architecture in which peptide–HLA presentation is first modeled and subsequently used to condition TCR recognition on the underlying peptide–HLA landscape (Fig. 1d, e), thereby enforcing biological consistency. To address the scarcity of validated negative TCR–pHLA pairs, we introduce dynamic and mutation-aware negative sampling to improve discrimination between true binders and near-neighbor decoys. Together, these architectural and training innovations enable StriMap to function as a unified framework for TCR or neoepitope prioritization in cancer immunotherapy and the identification of disease-associated antigens in autoimmunity (Fig. 1f), which we systematically evaluate in the following sections.

StriMap improves pHLA and TCR–pHLA benchmark performance

We evaluated StriMap against representative published state-of-the-art models on two related tasks—pHLA presentation prediction and TCR–pHLA interaction prediction—using benchmark datasets curated from independent prior studies with published train/test splits to enable reproducible comparison (Supplementary Tables 2 and 3). To provide a complete and transparent comparison, we re-benchmarked the included methods across all evaluated datasets whenever runnable implementations and compatible input requirements were available, using harmonized preprocessing, data splits, and evaluation metrics. Published results were used only when they corresponded to the same dataset split and evaluation protocol. We considered in-distribution (ID) testing, where test examples are drawn from the same underlying distribution as training; distribution-shifted (DS) testing across different studies or cohorts (dataset-level shift); and epitope-level out-of-distribution (OOD) generalization to epitopes not seen in training, forming an increasingly stringent hierarchy from ID to DS to OOD. For each benchmark, StriMap was trained or fine-tuned only on the corresponding training split unless otherwise stated, with evaluation performed on the held-out test split defined by the original study; dataset-specific positive and negative sample counts are summarized in Supplementary Data 1.

Across multiple pHLA presentation benchmarks^{12,13,22}, StriMap achieved top overall performance and remained robust under increasingly stringent ID/DS/OOD settings (Fig. 2a), with extended metric-level benchmarking results summarized in Supplementary Fig. 1. Performance gains were also consistent across individual HLA alleles: on the Que et al. dataset (ID)²², introduced alongside deepAntigen, StriMap improved per-allele AUPRC relative to deepAntigen²² (Fig. 2b), and broader per-allele comparisons against additional state-of-the-art predictors further supported consistent improvements across HLA classes (Supplementary Fig. 2). StriMap also consistently surpassed NetMHCpan¹¹ across all evaluations, underscoring robust gains over a widely used reference method in the literature. These gains translated to tumor neoantigen prioritization: following the immunogenicity fine-tuning workflow used by BigMHC, we fine-tuned StriMap on PRIME1.0/PRIME2.0 neoantigen immunogenicity data and evaluated the resulting StriMap-IM model on both the TESLA (Tumor Neoantigen Selection Alliance)³⁶ and CEDAR (Cancer Epitope Database and Analysis Resource)³⁷ benchmarks (Fig. 2c). StriMap-IM achieved top performance on CEDAR and ranked second on TESLA, while outperforming deepAntigen-IM, NetMHCpan11, and

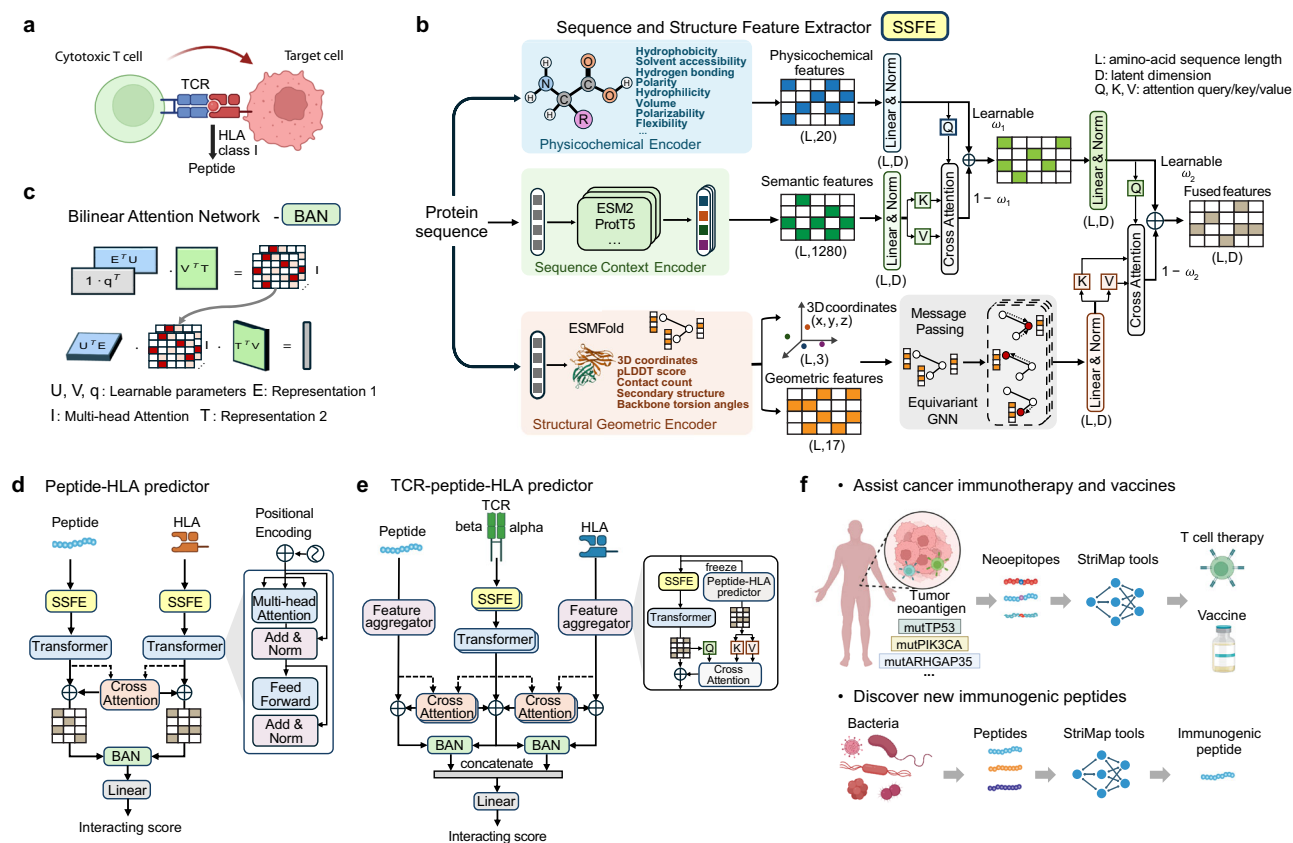


Fig. 1 | Overview of StriMap framework. **a** Biological context of T cell-mediated immune recognition. Cytotoxic T cells recognize target cells through the interaction between the T cell receptor (TCR) and peptide-HLA (pHLA) complexes presented on the surface of target cells. This tri-molecular interaction underlies antigen-specific immune surveillance and forms the biological basis of StriMap. **b** The architecture of the Sequence and Structure Feature Extractor (SSFE). Given a protein amino-acid sequence, SSFE integrates three complementary encoders: i) a physicochemical encoder that computes residue-level physicochemical features; ii) a sequence-based context encoder (e.g., ESM2 or ProtT5) that provides contextual sequence embeddings; and iii) a structural-derived geometric encoder (e.g., ESM-Fold) that generates 3D structural representations. Structural information is further processed through a multi-layer equivariant graph neural network (EGNN). All features are fused using gated cross-attention and layer-normalization to produce a unified representation. **c** The bilinear attention network (BAN) models pairwise interactions between two representations using learnable bilinear projections to generate multi-head attention maps. **d** Architecture of the peptide-HLA interaction

predictor. Peptide and HLA sequences are independently encoded by SSFE and processed through transformer blocks, followed by gated cross-attention and BAN fusion. The fused representation is fed into a linear layer to produce an interaction score between 0 and 1. **e** Architecture of the TCR-peptide-HLA interaction predictor. TCR α/β chains, peptide, and HLA sequences are independently encoded by SSFE and then passed through a feature aggregator module that incorporates and leverages the embedding representations learned from the peptide-HLA predictor. The aggregated features are further integrated via gated cross-attention and BAN, and the fused representation is mapped through a linear layer to produce an interaction score between 0 and 1. **f** Downstream applications of StriMap. StriMap enables systematic modeling of peptide-HLA and TCR-peptide-HLA interactions, facilitating the prioritization of immunogenic neoepitopes for cancer immunotherapy and vaccine development, as well as the discovery of pathogenic or immunodominant epitopes. Created in BioRender. Kang, H. (<https://BioRender.com/vhttfon>).

additional peptide-HLA presentation predictors across both benchmarks. These results support the utility of StriMap-derived representations for practical neoantigen prioritization, where only a limited number of candidates can be advanced for experimental testing.

To probe what StriMap learns beyond aggregate accuracy, we trained a pHLA model on a comprehensive dataset assembled from multiple sources. This dataset was allele-imbalanced (Supplementary Fig. 3). Visualization of the learned embedding space revealed a biologically meaningful organization, where the cluster separation reflected not only the individual HLA alleles but also the fine-grained heterogeneity within each allele, determined by distinct peptide-binding properties (Fig. 2d and Supplementary Fig. 4). For example, in HLA-B*27:05, peptides segregated into two major Leiden clusters (2 and 21) with distinct motifs: cluster 2 recapitulated the canonical preference for Arg at position 2 and Leu at the C terminus, whereas cluster 21 showed non-canonical preferences at position 2, indicating sub-allelic structure beyond a single consensus motif. Complementing embedding-level structure, attention-based analyses further revealed

allele-specific binding-position profiles and distinct interaction patterns between peptide residues and HLA contact residues, including differences between closely related alleles such as HLA-A*02:01 and HLA-A*02:03 (Supplementary Fig. 5). Finally, systematic saturation mutagenesis on a well-characterized influenza epitope (GILGFVFTL) demonstrated allele- and position-specific sensitivity in predicted binding scores (Supplementary Fig. 6). In particular, canonical anchor positions P2 and P9 showed significantly greater mutation-induced sensitivity than non-anchor positions across representative HLA alleles, supporting that StriMap captures biologically meaningful constraints underlying peptide-HLA binding.

We next evaluated StriMap on the more important task of TCR-pHLA interaction prediction across multiple independent benchmarks^{6,15–17,38}. Sequence similarity analysis confirmed that test TCRs across these datasets exhibit substantial divergence from the training data, ensuring the evaluation of true generalization rather than memorization of highly similar clonotypes (Supplementary Fig. 7). Given the rapidly expanding landscape of TCR specificity

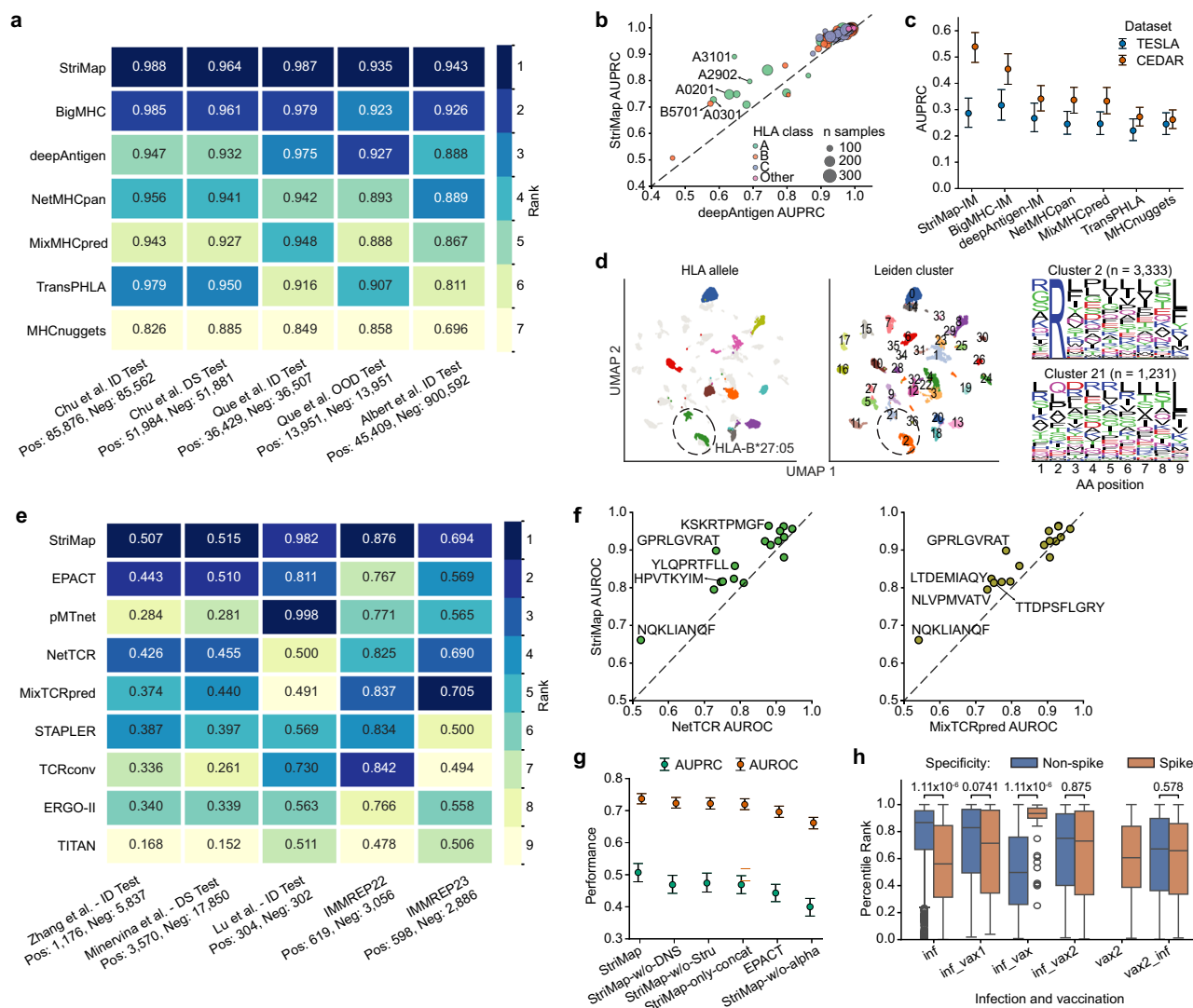


Fig. 2 | StriMap achieves robust prediction of pHLA presentation and TCR-pHLA recognition. **a** pHLA presentation prediction performance across benchmark datasets under ID (in distribution), DS (distribution shift), and OOD (out of distribution) settings. Values indicate AUPRC scores, and color intensity denotes method ranking within each dataset. Numbers of positive (Pos.) and negative (Neg.) samples are shown. **b** Comparison of StriMap and deepAntigen performance (AUPRC) across individual HLA alleles on the Que et al. dataset (ID). Dashed line indicates equal performance. **c** AUPRC comparison on TESLA and CEDAR neoantigen benchmarks. Points indicate median AUPRC values across bootstrap iterations ($n = 1000$), and error bars the 95% bootstrap confidence intervals. “IM” denotes models fine-tuned on PRIME1.0/PRIME2.0 neoantigen immunogenicity data following the BigMHC workflow. **d** UMAP of peptide embeddings, colored by 10 most abundant HLA alleles (left) and Leiden clusters (right). Two clusters (dashed ellipse) corresponding to HLA-B*27:05 exhibit divergent peptide-binding motifs (right). **e** TCR-pHLA interaction prediction performance across benchmark datasets under ID and DS settings. Values indicate AUPRC or benchmark-specific scores, and color intensity denotes method ranking within each dataset. Numbers

of positive (Pos.) and negative (Neg.) samples are shown. **f** Per-peptide AUROC comparison between StriMap and NetTCR (left) and MixTCRpred (right) on the IMMREP22 benchmark. Dashed line indicates equal performance. **g** AUROC and AUPRC comparison for StriMap and ablation variants under the same evaluation protocol. Points indicate median performance values across bootstrap iterations ($n = 1000$), and error bars the 95% bootstrap confidence intervals. StriMap-w/o-alpha excludes TCR α -chain input, StriMap-only-concat replaces cross-component interaction modules with simple concatenation, StriMap-w/o-Stru removes structural information, and StriMap-w/o-DNS removes dynamic negative sampling. **h** Percentile binding rank distributions of TCRs across SARS-CoV-2 cohorts from the Minervina et al. dataset. Box plots show the median (center line), 25th and 75th percentiles (box limits), and $1.5 \times$ the interquartile range (whiskers). Sample sizes (Non-spike/Spike TCRs): inf, $n = 1,164/76$; inf_vax1, $n = 303/137$; inf_vax, $n = 40/48$; inf_vax2, $n = 401/175$; and vax2_inf, $n = 962/172$. For vax2, only spike-specific TCRs were available ($n = 92$). P values (shown above plot), two-sided Mann-Whitney U tests with Benjamini-Hochberg.

prediction tools, we summarized representative prior methods and their key methodological characteristics in Supplementary Data 2, and then benchmarked StriMap against selected state-of-the-art models. Whenever runnable implementations and compatible input requirements were available, methods were re-run using harmonized pre-processing, dataset splits, and evaluation metrics; published results were used only when they matched the same split and evaluation protocol. When benchmark datasets included predefined positive and

negative pairs, we used the original labels directly to preserve the published evaluation protocol. For datasets lacking explicit negative pairs, we generated negatives by random TCR and pHLA shuffling while excluding known positives, consistent with prior benchmarking practices³⁹. We additionally incorporated dynamic negative sampling (DNS), regenerating negatives each epoch of model training to expand the effective negative space.

Overall, StriMap achieved strong performance under both ID and DS settings (Fig. 2e), with per-peptide comparisons on Zhang et al. (ID), Minervina et al. (DS) and IMMREP22 datasets (Supplementary Figs. 8–10); Fig. 2f highlights the IMMREP22 per-peptide AUROC comparisons, where StriMap outperformed NetTCR¹⁸ and MixTCRpred²⁴, two widely used reference methods. We note that the Lu et al. benchmark¹⁷ adopts peptide-permutation negatives that preserve TCR–HLA pairs but scramble peptide identity (Supplementary Fig. 11). Under this scheme, negatives can be separated largely via peptide–HLA compatibility rather than true TCR-specific recognition, potentially inflating performance. Because StriMap jointly models pHLA presentation and TCR–pHLA recognition, it can exploit peptide–HLA compatibility signals and achieves substantially higher performance under this protocol. In contrast, methods that ignore HLA information or model TCR–pHLA interactions in isolation are structurally disadvantaged.

To assess the contribution of paired TCR-chain information, we first removed the TCR α -chain input while keeping the same training and evaluation protocol on Zhang et al. dataset. This reduced AUROC from 0.737 to 0.662 and AUPRC from 0.507 to 0.400, indicating that the α -chain provides substantial specificity information beyond the β -chain alone. Ablation analyses further showed that removing the structural module, DNS, or replacing the cross-component cross-attention/BAN interaction modules with simple concatenation degraded AUROC/AUPRC relative to the full model (Fig. 2g). Beyond ablation analysis, peptide-length-stratified evaluations showed that StriMap supports variable-length peptide inputs and maintained robust performance across the evaluated peptide-length range in both pHLA and TCR–pHLA benchmarks (Supplementary Fig. 12a, b). SHAP analysis further quantified fine-grained feature importance at both the modality and component levels, identifying sequence and pHLA contexts as the primary drivers of predictions, while physicochemical and structural features provided meaningful contributions that validate our multi-modal design (Supplementary Fig. 12c, d). Notably, the integrated pHLA context exhibited a dominant SHAP score compared to isolated peptide and HLA features. This reflects our architecture: the fused pHLA context captures their synergistic functional dynamics, naturally absorbing their individual predictive weights during the attribution process.

To assess generalization across clinical cohorts, we examined the SARS-CoV-2-related TCR–pHLA interactions in the Minervina et al. dataset³⁸, comprising 3570 TCRs across 14 peptides (5 spike and 9 non-spike). Cohorts included recovered individuals' post-infection (inf), recovered after one vaccine dose (inf_vax1), sampled immediately after two doses (inf_vax), post-vaccination recovery (inf_vax2), vaccinated naïve individuals (vax2), and breakthrough infections (vax2_inf) (Fig. 2h). We predicted the TCR binding score for each spike or non-spike peptide across all categories, and summarized the results as percentile binding rank distributions (Fig. 2h). Within each category, spike- and non-spike-specific TCRs were compared using two-sided Mann–Whitney *U* tests with Benjamini–Hochberg correction. Vaccination-associated cohorts (“vax”) are expected to mount elevated spike-specific T cell responses following spike immunization; indeed, we found more favorable predicted binding ranks for spike-specific TCRs. This effect was most pronounced in the inf_vax cohort, where spike-specific TCRs achieved significantly higher predicted binding-rank percentiles than non-spike-specific TCRs within the same cohort (adjusted $P < 1 \times 10^{-4}$), consistent with recent vaccination driving expansion and activation of spike-specific clonotypes. In contrast, at later time points (inf_vax2) and in breakthrough infections (vax2_inf), this separation was attenuated and was not statistically significant, likely reflecting contraction of vaccine-induced responses and diversification of the T cell repertoire following additional antigen exposures.

We further assessed StriMap on the IMMREP23 challenge¹⁶, where participants could incorporate external training data beyond the official dataset, and the exact data sources used by different methods are not fully disclosed. We compared three training strategies: (i) training only on the organizer-provided data, (ii) training on a comprehensive dataset assembled from diverse public studies^{15,16,23,40–42} (Supplementary Fig. 13), and (iii) an ensemble combining multiple negative sampling strategies (Supplementary Fig. 14a). With the ensemble strategy, StriMap ranked third with AUC0.1 = 0.6943 (Supplementary Fig. 14b; a definition of the AUC0.1 is provided in the Methods). Notably, the top two methods (IMW-DETECT and MixTCRpred_s2) leveraged undisclosed in-house data, complicating attribution of gains to architecture versus data. We further evaluated StriMap in a zero-shot (OOD) setting with test epitopes absent from training; despite only three unseen epitopes, performance exceeded random baselines, indicating partial generalization, but remained limited, potentially due to substantial epitope divergence (Supplementary Fig. 14c). These results suggest that few-shot learning is often more reliable in practice, while still motivating downstream use cases such as cancer TCR and neoepitope prioritization.

TCR and neoepitope prioritization for cancer immunotherapy

Somatic mutations in cancer can generate neoepitopes that are processed and presented by HLA molecules and can be recognized by T cells (Fig. 3a), providing a key rationale for T cell-based immunotherapies. To demonstrate the translational utility, we applied StriMap to two clinically relevant cancer immunotherapy tasks: (i) TCR-centric prioritization for recurrent mutant peptide–HLA pairs in the context of adoptive T cell therapy, and (ii) neoepitope prioritization for personalized cancer vaccine design (Fig. 3b). These two settings reflect common decision points in T cell-based immunotherapy, where only a small number of candidates can be experimentally evaluated. For the cancer prioritization analyses, we used StriMap trained on the comprehensive TCR–pHLA dataset (Supplementary Fig. 13), without task-specific fine-tuning; positive/negative sample counts are summarized in Supplementary Data 1.

We first considered task 1, a TCR-centric setting in which recurrent cancer-associated mutations are known, and the goal is to identify cognate TCRs suitable for adoptive cell therapy. Among these, mutations in TP53 represent one of the most frequent and well-characterized classes of cancer driver alterations across tumor types, and several TP53-derived mutant peptides have been shown to elicit T cell responses in patients⁴³. Using representative TP53 hotspot mutations with experimentally observed cognate TCRs originally evaluated by TAPIR²⁵ (Fig. 3c and Supplementary Data 3), we evaluated StriMap's ability to rank antigen-specific TCRs for mutant peptide–HLA queries under a fully zero-shot (OOD) setting, where neither the peptide nor the TCRs were seen during training. All positives and negatives were taken directly from TAPIR (7 cognate TCRs; 3144 negatives), and we compared against TAPIR and the NetTCR-2.0¹⁸ baseline as reported in the original study²⁵.

Within the TCR-centric ranking task, we aimed to resolve mutation-specific effects, where mutant peptides differ from their wild-type counterparts by only one or a few amino acids. Distinguishing such subtle differences likely requires amino-acid-level modeling rather than treating non-cognate pairs as the only negatives. To better reflect this biological setting, we introduced a mutation-aware negative sampling strategy, generating negatives via random amino acid substitutions in either the peptide or the TCR CDR3 α/β regions. Across both AUROC and AUPRC for TCR ranking, StriMap variants outperformed TAPIR, NetTCR-2.0, and TAPIR-plus (trained with proprietary in-house TCR functional data not accessible to StriMap), with StriMap-M2 (peptide-mutation negatives) achieving the strongest overall performance and StriMap-M1 (TCR-mutation negatives) providing intermediate gains relative to the base model (Fig. 3d).

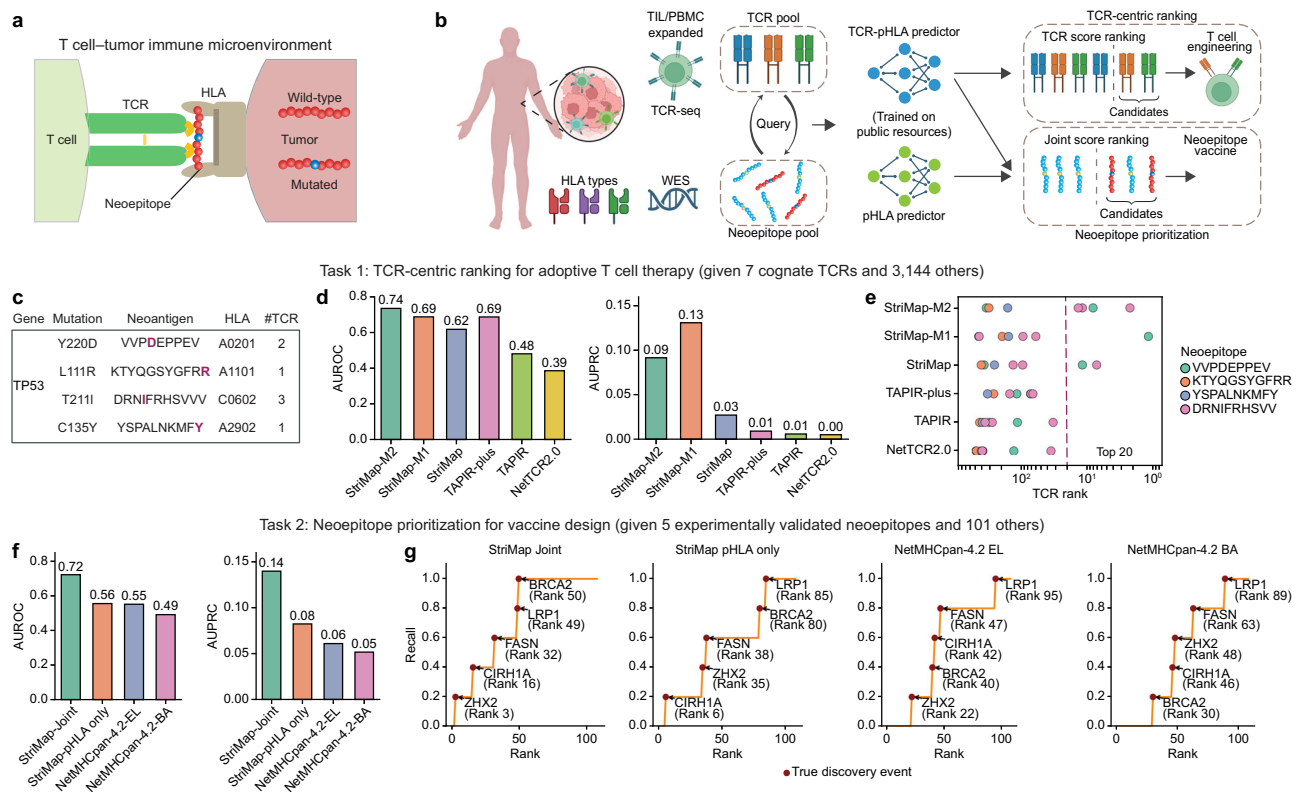


Fig. 3 | StriMap supports two clinically relevant tasks for T cell-based immunotherapy. **a** Schematic illustration of T cell recognition within the tumor immune microenvironment, highlighting interactions between TCRs and neoepitope-HLA complexes presented on tumor cells. **b** Overview of the StriMap framework for joint analysis of HLA presentation and TCR recognition. Whole-exome sequencing (WES), HLA typing, and TCR sequencing define candidate neoepitopes and patient-specific TCR pools, which are evaluated using StriMap pHLA and TCR-pHLA predictors for downstream prioritization. **c** Representative TP53 cancer mutations, their corresponding neoepitopes, and the number of experimentally observed TCRs recognizing each neoepitope. **d** Performance comparison of StriMap with TAPIR and NetTCR for prioritizing cognate TCRs given mutant pHLA queries, evaluated by AUROC and AUPRC. TAPIR-plus denotes TAPIR trained on both public datasets and additional in-house proprietary TCR functional data, which StriMap does not access. StriMap-M2 indicates fine-tuning using negative samples generated by randomly mutating one to two amino acids in the peptide, whereas StriMap-M1 indicates fine-tuning using negative samples generated by randomly

mutating one to three amino acids in the TCR CDR3 α and/or CDR3 β regions. For reference, the random baseline is AUROC = 0.5 and AUPRC = 0.002. **e** TCR ranking results for TP53-derived neoepitopes. Each point represents a TCR-pHLA interaction for a given neoepitope. Colors denote different neoepitopes, and the dashed line highlights the top 20 ranked TCR candidates. **f** AUROC and AUPRC performance of StriMap for prioritizing immunogenic neoepitopes given melanoma patient-specific TCR pools, using a joint score combining TCR-pHLA and pHLA predictors, compared with the pHLA-only StriMap model and NetMHCpan-4.2-EL and NetMHCpan-4.2-BA. For reference, the random baseline is AUROC = 0.5 and AUPRC = 0.047. **g** Recall-rank curves illustrating the recovery of true immunogenic neoepitopes across ranking positions for TCR involved predictions versus pHLA predictor only predictions. Red markers denote true discovery events. NetMHCpan-4.2 EL and BA denote the eluted-ligand and binding-affinity prediction modes, respectively. Panels a–b were created in BioRender. Kang, H. (<https://BioRender.com/qeveyxr>).

To determine if this mutation-aware sampling strategy is model-agnostic, we applied it to baseline models (EPACT-M2 and NetTCR-M2). While both exhibited substantial AUROC boosts (Supplementary Fig. 15), a distinct architectural dependency emerged: complex, Transformer-based models (StriMap and EPACT) benefited profoundly in AUPRC, whereas simpler network architectures like NetTCR failed to show AUPRC improvements. This suggests that models with lower representational capacity struggle to fully learn the fine-grained interaction shifts induced by single-amino-acid mutations. These results indicate that mutation-aware negative sampling is particularly beneficial for modeling cancer-associated antigen recognition.

Consistent with aggregate performance metrics, StriMap showed improved enrichment of true binders among the highest-ranked TCR candidates relative to competing methods (Fig. 3e). For example, multiple experimentally validated TCRs recognizing neoepitopes DRNIFRHSVV and VVPDEPPEV were ranked within the top 20 by StriMap, whereas other methods more often ranked these true positives substantially lower. This capability is critical for T cell therapy, where only a small number of top-ranked candidates can be advanced for costly experimental validation and clinical development. We next

asked whether StriMap can resolve mutation-specific effects for a fixed TCR by comparing mutant and wild-type peptide-HLA contexts. Because wild-type peptides may lack immunogenicity either due to limited TCR recognition or low HLA presentation, this comparison helps disentangle the underlying driver. Notably, predicted differences were driven predominantly by changes in pHLA presentation rather than large shifts in inferred TCR–pHLA binding scores (Supplementary Fig. 16). For example, for the C135Y mutation, the predicted HLA presentation score increased from 0.06 (wild-type) to 0.65 (mutant), whereas the inferred TCR–pHLA score changed only modestly from 0.61 to 0.65, suggesting that the gain in immunogenicity is primarily consistent with enhanced HLA presentation. Together, these results illustrate how jointly modeling pHLA presentation and TCR recognition enables StriMap to predict whether mutation-driven effects are more consistent with altered HLA presentation or altered TCR recognition.

We next evaluated StriMap in task 2, a complementary neoepitope-centric setting, where patient-specific TCR repertoires are available, and the objective is to prioritize immunogenic neoepitopes for vaccine development (Fig. 3b). We used paired neoantigen and TCR

repertoire data from a melanoma dataset², including clonally expanded TCRs profiled from tumor-infiltrating lymphocytes (TILs) and matched peripheral blood mononuclear cells (PBMCs). In total, this dataset comprised 106 candidate peptide mutations, of which 5 were experimentally confirmed to elicit T cell immune responses. Candidate neopeptides were ranked using a joint score that integrates predicted neopeptide-HLA presentation with predicted TCR interaction score across the TCR repertoire; we refer to this joint scoring framework as StriMap-joint. Using the mutation-aware StriMap-M2 model, we summarized each candidate neopeptide by the mean interaction score of the top 30 ranked TCR–pHLA pairs, capturing the collective recognition potential of the patient’s expanded T cell repertoire. StriMap-joint consistently achieved higher AUROC and AUPRC than both a pHLA-only StriMap model and NetMHCpan-4.2-based baselines (EL and BA) used in the original study², highlighting the benefit of explicitly incorporating TCR recognition into neopeptide prioritization (Fig. 3f). This advantage was robust to the choice of the number of top-ranked TCR–pHLA pairs used for aggregation (Supplementary Fig. 17). Recall-rank analysis, summarizing the ranking positions of all experimentally validated immunogenic neopeptides within the ranked candidate list, further showed that true immunogenic neopeptides were placed earlier in the ranking when using the joint model compared with pHLA-only or NetMHCpan-based approaches (Fig. 3g).

Together, these results indicate that StriMap’s joint modeling of peptide presentation and TCR recognition can support zero-shot prioritization in two cancer immunotherapy settings. By capturing mutation-driven changes in pHLA presentation and their downstream impact on TCR engagement, StriMap provides a unified framework for TCR selection in adoptive cell therapy and neopeptide ranking for personalized cancer vaccines.

Proteome-wide discovery of bacterial mimic peptides in AS

Having established StriMap’s utility in cancer immunotherapy, we next asked whether the same joint modeling of peptide–HLA presentation and TCR recognition could be extended to autoimmune T cell recognition, where the relevant antigenic triggers are typically unknown. We focused on ankylosing spondylitis (AS), a prototypical HLA-B27-associated inflammatory disease, motivated by prior evidence that AS is strongly linked to HLA-B27 and that disease-associated TCRs enriched for a conserved TRBV9 motif may contribute to pathogenesis. A long-standing question in AS is how environmental exposures, particularly gut microbial antigens, may interact with HLA-B27 to initiate or amplify autoreactive T cell responses. Molecular mimicry provides one possible mechanism: microbial peptides presented by HLA-B27 may activate TCRs that also cross-react with self-antigens, thereby linking bacterial antigen exposure to autoimmune inflammation. We therefore focused on bacterial peptides not as arbitrary foreign antigens, but as candidate environmental triggers that could be presented by HLA-B27 and recognized by AS-associated TRBV9 TCRs. To define a biologically grounded and clinically relevant search space, we curated a literature-supported cohort of AS-associated gut bacterial strains spanning four major phyla: *Bacillota/Firmicutes*, *Verrucomicrobiota*, *Bacteroidetes*, and *Proteobacteria*. The final panel comprised 16 strains, totaling 43,241 annotated proteins used for downstream peptide enumeration (Supplementary Table 4). This design allowed us to ask whether StriMap could prioritize microbial peptides with both predicted HLA-B27 presentation and predicted recognition by disease-associated TCRs from a proteome-scale bacterial search space.

To evaluate whether StriMap can generalize across patients in a realistic translational regime, we designed a cross-patient few-shot inference workflow centered on TRBV9 TCRs (Fig. 4a). Specifically, we used TCR binding data from patient 1 (TRBV9 TCR AS3.1) and patient 2 (TRBV9 TCRs AS4.1–AS4.4) as reference inputs and then predicted peptide-binding specificity for a query TRBV9 TCR (AS8.4) from an

independent patient 3. AS8.4 was selected because it was not among the five TRBV9 TCRs used for few-shot training⁴⁴, enabling an out-of-sample evaluation, and because it showed the strongest activation to the positive-control peptides among the remaining TRBV9 TCRs tested in the original study⁴⁴. This design simulates deployment on a new patient with limited disease-specific supervision while still leveraging large public training resources (Supplementary Data 1). To examine whether the trained model relied on biologically plausible recognition determinants in this setting, we applied residue-level attribution analysis to experimentally solved AS-associated TCR–YElH–HLA-B*27:05 complex structures, including AS4.2, AS4.3, and AS8.4 (Supplementary Fig. 18). This analysis showed that high-importance residues were concentrated in CDR3 regions of both TCR α and TCR β chains, supporting that StriMap captures biologically relevant TCR recognition regions rather than broadly distributed sequence signals.

We then performed proteome-wide *in silico* screening of bacterial peptides at the scale required for mimicry discovery (Fig. 4b). For each bacterial proteome, we generated overlapping 9-mer peptides using a sliding window ($k = 9$, stride = 1), yielding a pool of 13,068,902 candidate 9-mers. StriMap was then applied in a two-stage, dual-task manner: (i) predicting the score of HLA-B*27:05 presentation, followed by (ii) predicting recognition by the disease-associated TRBV9 TCRs, thereby coupling presentation and recognition in a single screening workflow. In our implementation, we retained the top predicted HLA-B*27:05 binders (top 3% relative to a random-peptide background; 568,982 peptides) for downstream TCR screening (Fig. 4c). After filtering for high-confidence HLA-B*27:05 candidates, we used the StriMap TCR–pHLA predictor to estimate interactions between each candidate peptide and five disease-associated TRBV9 TCRs (AS3.1, AS4.1–AS4.4) previously reported to be reactive to multiple HLA-B*27:05-bound peptides⁴⁴. Peptides were then ranked by the mean predicted binding score across the five TCRs, and we selected seven top-ranking bacterial peptides that were each predicted to interact strongly with at least two TRBV9 TCRs. We imposed this multi-TCR criterion to reflect the known cross-reactivity of disease-associated TRBV9 TCRs and to prioritize candidates with broader potential to activate multiple TRBV9 clonotypes instead of peptides specific to a single TCR, thereby increasing translational relevance. We next evaluated the predicted reactivity of these candidates against the out-of-sample query TRBV9 TCR AS8.4. We benchmarked the candidate bacterial peptides against negative controls and two previously reported positive controls: the human self-peptide TRLALIAPK and the bacterial mimic peptide LRVMLAPF. Furthermore, to rigorously test the model’s discriminative power, we included human-derived homologous peptides—identified via BLAST for their high linear sequence similarity to the bacterial sequences in our screening pool—as stringent specificity checks to demonstrate StriMap’s ability to distinguish true structural mimics from mere sequence neighbors. Top-ranked candidates included peptides derived from *Streptococcus* spp. and *A. muciniphila*, and multiple high-scoring candidates (e.g., ARVMALMPF, GRILALVPK, PRIMMISMK, PRLPLTVPF) emerged among the leading predictions under this cross-patient few-shot setting (Fig. 4d). The main conclusion from this computational screen is that StriMap can reduce a large disease-associated microbial proteome space to a small set of candidate HLA-B27-presented bacterial peptides predicted to engage AS-associated TRBV9 TCRs.

Experimental validation of TRBV9⁺ T cell activation

To experimentally validate StriMap-prioritized candidate mimic peptides in an HLA-B*27:05-restricted setting, we implemented a discovery workflow in which computational screening nominates candidate TCR–peptide–HLA interactions for targeted testing and iterative refinement (Fig. 5a). Unlike prior validation settings that primarily tested known epitopes, binding-level readouts, or bulk peptide immunogenicity, this workflow combines proteome-scale discovery

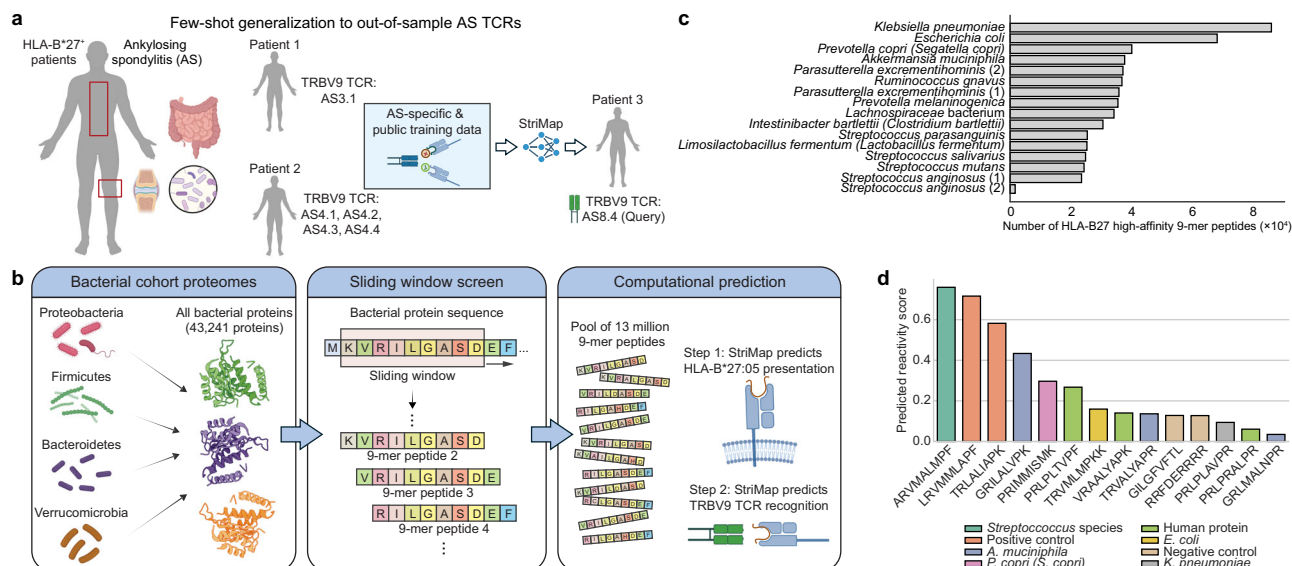


Fig. 4 | Computational discovery of AS-associated bacterial mimicry peptides through cross-patient prediction. **a** AS-associated bacterial cohort and cross-patient few-shot prediction setup. We curated a literature-derived cohort of gut bacterial strains previously reported to be associated with ankylosing spondylitis (AS), spanning four major phyla: Bacillota, including *Streptococcus* spp., *R. gnavus*, etc.), Verrucomicrobiota (*A. muciniphila*), Bacteroidetes (*P. copri*, *P. melaninogenica*), and Proteobacteria (*K. pneumoniae*, *E. coli*, *Parasutterella excrementihominis*). To assess generalization to unseen patient TCRs, we implemented a cross-patient inference workflow in which TRBV9 TCR binding data from patient 1 (AS3.1) and patient 2 (AS4.1–AS4.4) were used as reference inputs to train the specificity model, and StriMap then predicted peptide-binding specificity for a query TRBV9 TCR from an independent Patient 3 (AS8.4), simulating a few-shot setting. **b** Large-scale in silico 9-mer peptide screening pipeline. The proteomes of the selected bacterial cohort, comprising 43,241 proteins, were retrieved and processed. A sliding window approach with a window size of $k = 9$ and a stride of 1 amino acid was applied to

generate a pool of ~13 million 9-mer peptides. The StriMap computational framework performs a dual-task prediction: Task 1 predicts the likelihood of HLA-B*27:05 presentation, and Task 2 predicts the recognition probability by specific TRBV9 TCRs (e.g., AS8.4). **c** Landscape of HLA-B27 high-affinity 9-mer peptides across the cohort. The bar chart illustrates the distribution and total count of predicted high-affinity 9-mer peptides for each bacterial strain. Strains such as *K. pneumoniae* and *E. coli* exhibit a high density of potential HLA-B27 binders, highlighting their significance in the molecular mimicry hypothesis. **d** Predicted reactivity scores for candidate bacterial peptides. Representative peptides from various phyla were ranked by their predicted reactivity scores against the target TCR. High-scoring candidates from *Streptococcus* spp. and *A. muciniphila* are compared against positive and negative controls, indicating potential cross-reactivity between bacterial antigens and AS-associated TCRs. **a**, **b** were created in BioRender. Kang, H. (<https://BioRender.com/77gafpz>).

with TCR-defined functional activation assays for newly predicted bacterial mimic peptides (Supplementary Table 5). We generated endogenous TCR-knockout Jurkat cell lines and re-expressed the TRBV9 TCR AS8.4 by lentiviral transduction (“Materials and Methods”). Antigen-presenting cells (APCs) were generated by transducing 293 T cells to express HLA-B*27:05. T cells were then co-cultured with APCs that had been pre-incubated with the candidate peptides, and T cell activation was assessed by measuring both CD69 surface expression and fluorescence from the NFAT-driven ZsGreen reporter, in which ZsGreen expression is controlled by a nuclear factor of activated T cells (NFAT) response element to report downstream TCR signaling (Fig. 5b; “Materials and Methods”).

Although not all candidates were predicted to bind AS8.4, we tested the full set of 10 peptides to directly assess the model’s ability to separate activators (CD69⁺ZsGreen⁺) from non-binders. We observed robust T cell activation in response to the three predicted positive binders: ARVMALMPF (from *Streptococcus anginosus*, *S. mutans*, and *S. parasanguinis*; 49.5%), PRIMMISMK (from *Prevotella copri* and *Segatella copri*; 1.29%), and GRILALPK (from *Akkermansia muciniphila*; 0.97%) (Fig. 5c and Supplementary Fig. 19). For comparison, we included two previously reported positive controls (TRLALIPK, 7.0%; LRVMMLAPF, 57.8%), and two negative controls (RRFDERRPR, 0.057%; GILGFVFTL, 0.084%) together with background conditions (no peptide; no APC/no peptide), all of which showed minimal activation.

StriMap predictions across the ten peptides were highly concordant with experimental outcomes: the three activating peptides (ARVMALMPF, GRILALPK, PRIMMISMK) were ranked among the top candidates, whereas non-activating peptides received consistently low

scores. To contextualize these results, we benchmarked multiple modeling strategies on the experimentally tested peptide–TCR pairs by labeling the three activating peptides as positives and the remaining peptides as negatives, and computing AUROC/AUPRC from each method’s predicted scores (Supplementary Fig. 20). StriMap substantially outperformed logistic regression, the amino-acid frequency–based baselines used in Yang et al.⁴⁴, as well as NetTCR-2.2¹⁹, in both metrics. Notably, incorporating limited experimental feedback in a few-shot setting improved performance, indicating that StriMap can efficiently leverage sparse experimental validation data to refine disease-specific TCR-antigen recognition predictions.

Finally, given reports linking potentially pathogenic *Streptococcus* species to AS, IBD, and AS–IBD overlap^{45,46}, we asked whether the experimentally validated candidates show broader clinical associations. We performed an exact k-mer search against a translated, non-redundant microbial protein catalog constructed from metagenomic mix-assemblies⁴⁷. This protein catalog was built from 156 individuals with IBD including ulcerative colitis (UC, $n = 29$) and Crohn’s disease (CD, $n = 82$) as well as from healthy controls ($n = 45$) across IBD cohorts^{48–50}, healthy controls from the Men’s Lifestyle Validation Study (MLVS) study ($n = 361$)⁵¹, and 970 patients with pathogen-confirmed gastroenteritis⁵². Matched metatranscriptomic data were available for all metagenomic samples, allowing us to evaluate the transcriptional activity of the genes encoding proteins that contain the candidate peptides. Across the ten candidate peptides, five proteins containing the ARVMALMPF motif were significantly more prevalent and enriched in patients with CD or UC compared to healthy controls, whereas two proteins with PRIMMISMK were more prevalent and transcriptionally

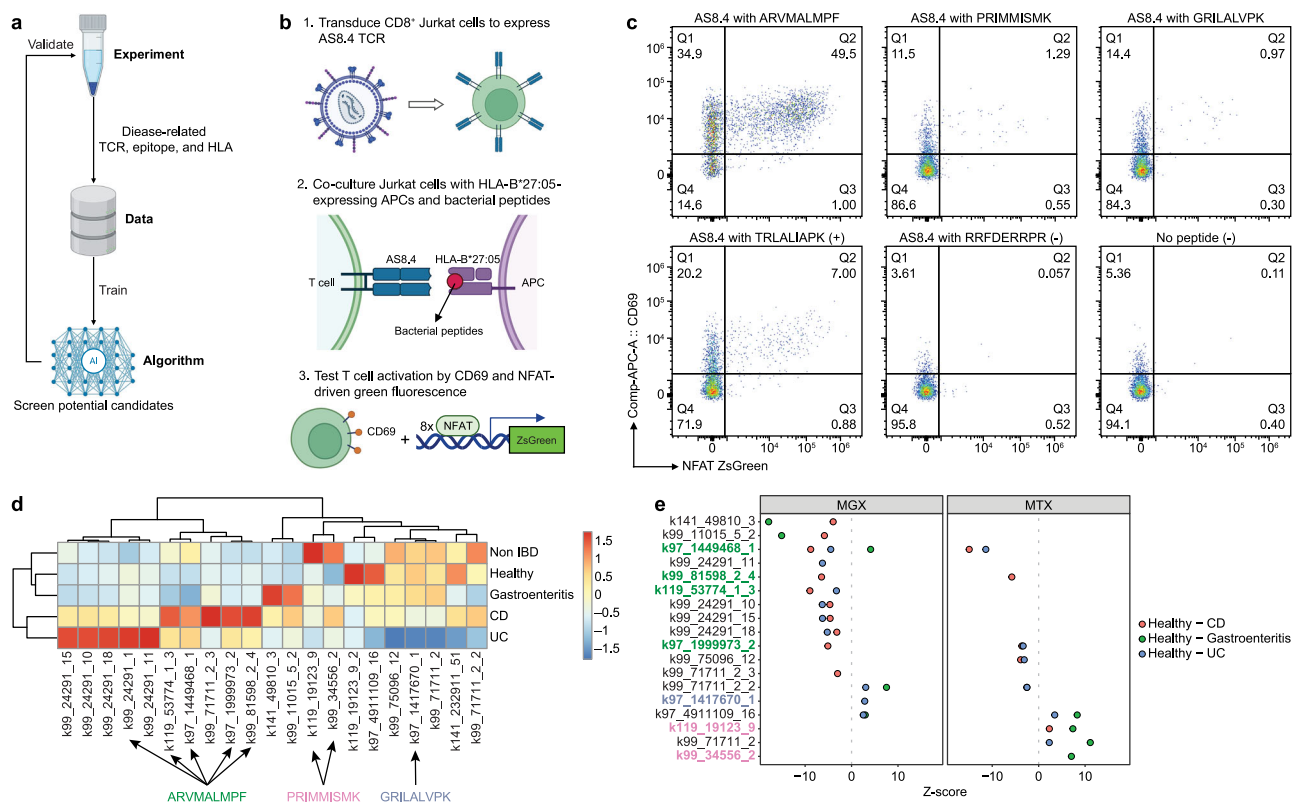


Fig. 5 | Experimental validation of AS-associated microbial mimicry peptides.

a Overview of discovery framework, in which StriMap prioritizes candidate disease-associated TCR-peptide-HLA interactions for experimental validation and iterative refinement. **b** Experimental workflow for validating interactions between the AS-associated TCR AS8.4 and candidate peptides. CD8⁺ Jurkat T cells were transduced to express the AS8.4 TCR and co-cultured with HLA-B*27:05-expressing antigen-presenting cells (APCs) pulsed with candidate bacterial peptides. **c** Flow cytometry analysis of T cell activation following peptide stimulation. Activation was quantified by co-expression of CD69 (y-axis) and NFAT (nuclear factor of activated T cells)-driven ZsGreen reporter activity (x-axis). Three bacterial peptides induced robust T cell activation, comparable to the positive control, whereas negative and background controls showed minimal activation. Activated cells are enriched in the CD69⁺ZsGreen⁺ quadrant. Representative flow-cytometry plots from one of two independent experiments are shown; both experiments produced consistent activation patterns. **d** Metagenomic and metatranscriptomic profiling of bacterial peptide candidates across clinical cohorts. Heatmap shows the relative prevalence of peptide-associated features in healthy controls, infectious gastroenteritis, Crohn's disease CD, and ulcerative colitis UC, following Z-score normalization. **e** Statistical comparison of peptide-associated features across diagnostic groups. A Kruskal-Wallis test was performed across healthy, infectious gastroenteritis, UC, and CD cohorts. Features with significant group-level differences ($p < 0.05$) were further assessed using post hoc Dunn's tests with multiple-comparison correction. Exact P values and Bonferroni-adjusted P values for all pairwise comparisons are provided in Supplementary Table 6. MGX Metagenomics; MTX Metatranscriptomics. **a**, **b** were created in BioRender. Kang, H. (<https://BioRender.com/gu35ufi>).

upregulated in non-IBD samples (Fig. 5d, e and Supplementary Table 6). In particular, ARVMALMPF—predicted with the highest score in the screening stage (Fig. 4d) and validated here to activate AS8.4⁺ T cells—showed consistent enrichment in IBD cohorts, supporting the possibility of shared microbial triggers between AS and IBD.

Discussion

Deep learning has enabled major progress across biomedicine, yet generalizing to new biological contexts remains challenging, particularly under data sparsity, class imbalance, and dataset bias. Rather than relying solely on ever-larger datasets, we focus on developing models that better capture biological structure under limited supervision, and apply this principle to TCR antigen-specificity prediction, a central problem in infection, cancer, and autoimmunity. T cell-mediated immunity relies on a recognition cascade in which peptides are presented by HLA molecules and subsequently engaged by diverse T cell receptors. Modeling this tri-molecular process has been difficult because the interaction space is combinatorially large, recognition is structurally constrained yet flexible, and available experimental data are sparse and biased. Here we present StriMap, a unified framework that jointly models peptide-HLA presentation and TCR recognition

using complementary physicochemical, sequence-context, and structure-informed representations.

A central conceptual advance of StriMap is its treatment of antigen presentation and TCR recognition as a coupled biological process rather than independent prediction tasks. By conditioning TCR recognition on the peptide-HLA landscape, StriMap provides a biologically consistent decision pathway that helps disentangle whether changes in immunogenicity arise from altered presentation or altered TCR engagement. Moreover, StriMap incorporates structure-derived constraints at scale without requiring experimentally solved complexes, enabling reasoning about interface geometry beyond current structural databases. In practice, we find that training strategy is a critical complement to architecture: dynamic and mutation-aware negative sampling exposes the model to realistic near-miss decoys, which is particularly important in clinical settings where neoantigens often differ from self by a single amino acid.

StriMap's utility is highlighted by applications in two distinct disease settings where candidate selection must be aggressively prioritized. In cancer immunotherapy, StriMap improves enrichment of true binders among the top-ranked candidates under zero-shot conditions, a setting that mirrors clinical reality because patient-specific neoantigens are inherently novel. The coupled modeling of

presentation and recognition provides a practical framework for prioritizing TCRs and vaccine neoepitopes when only a small number of candidates can be experimentally tested. Although we focus on melanoma and TP53 mutations here, we anticipate that this prioritization strategy may generalize to other cancer contexts. In autoimmunity, StriMap links a large-scale microbial sequence space to antigen-specific T cell responses: by combining disease-associated TRBV9 TCR clonotypes with an HLA-B*27:05-restricted microbial peptide landscape, we identified bacterial peptides that were experimentally validated to activate AS-associated T cells. The enrichment of one validated peptide in IBD cohorts further suggests that shared microbial antigenic drivers may contribute to overlapping inflammatory phenotypes, illustrating how StriMap can convert large-scale in silico screening into testable mechanistic hypotheses. Importantly, functional molecular mimicry often relies on shared biochemical hotspots rather than rigid global structural homology⁵³. StriMap successfully captures these non-linear geometric constraints, enabling the identification of diverse bacterial mimics that transcend simple sequence or structural alignment.

We envision StriMap as a task-specific predictive framework designed to rapidly evaluate large peptide–HLA and TCR–pHLA candidate spaces while explicitly coupling antigen presentation and TCR recognition. From a practical clinical perspective, StriMap can complement other structure modeling approaches, such as AlphaFold3⁵⁴ in a two-stage workflow. In this workflow, StriMap is first applied to prioritize a manageable set of candidate neoepitopes or TCR–pHLA pairs, after which structural modeling approaches are used to generate mechanistic hypotheses about how the selected targets interact.

Several limitations and opportunities remain. First, because TCR–pHLA data are sparse and unevenly distributed across epitope space, fully zero-shot generalization to entirely novel epitopes remains challenging⁵⁵. Our results, therefore, suggest that evaluation of zero-shot performance should emphasize functional prioritization, rather than relying solely on aggregate metrics such as AUROC. OOD benchmarks such as IMMREP23 include only a small number of unseen epitopes ($n=3$), limiting the interpretability of global performance statistics. In contrast, StriMap demonstrates strong practical utility in clinically relevant ranking settings: it placed validated TP53-specific TCRs among the top-ranked candidates in cancer immunotherapy, and prioritized a small set of activating peptides from a pool of ~13 million bacterial 9-mers in autoimmunity. Such top-K precision is often the key determinant of translational success, because it directly governs which candidates are selected for costly experimental validation. Nevertheless, few-shot adaptation using limited task- or patient-specific data remains a reliable strategy for further improving accuracy in applications requiring higher fidelity.

Second, structure-informed representations improve interface modeling but may not capture conformational dynamics that influence recognition; incorporating structural uncertainty (e.g., ensemble predictions) or dynamics-inspired features may further improve predictions. Consistent with this direction, the ongoing IMMREP25 Kaggle challenge underscores the importance of structure-aware TCR–pHLA modeling, although direct comparison is currently limited because top-performing structure-based approaches have not yet been published. Moreover, many leading models are based on AlphaFold3⁵⁴ prediction, whose computational cost and throughput constraints limit applicability to proteome-scale screening. StriMap circumvents this by modeling interfaces implicitly and employing a two-stage funnel strategy: relying on rapid sequence encoders for massive-scale pHLA screening, while engaging ESMFold-derived monomer structures specifically for the complex task of TCR recognition. This deliberate trade-off enables efficient prioritization across millions of candidates. Third, our current implementation focuses on HLA class I and CD8⁺ T cell recognition; extending the framework to HLA class II

presentation and CD4⁺ T cell recognition will be important for broadening translational impact.

To facilitate broad adoption and iterative improvement, we provide a web portal and training interface that support both prediction and rapid fine-tuning on local datasets, enabling users to incorporate sparse experimental feedback while retaining knowledge from large-scale pretraining (Supplementary Fig. 21). We anticipate that this combination of coupled modeling, structure-aware representations, and lab-in-the-loop validation will help bridge computational prediction and mechanistic immunology, enabling more effective discovery in cancer immunotherapy and antigen-driven autoimmune disease.

Methods

The StriMap framework

StriMap is a unified deep learning framework for modeling pHLA binding and TCR recognition of pHLA complexes at residue resolution. The framework consists of two modular components: (i) a peptide–HLA predictor that learns transferable pHLA representations, and (ii) a TCR–pHLA predictor that leverages pretrained pHLA features. Both components are built upon a unified Sequence and Structure Feature Extractor (SSFE) and attention-based residue-level interaction modeling using Bilinear Attention Networks (BAN).

Sequence and structure feature extractor (SSFE)

Given an input protein sequence of length L , SSFE constructs residue-level representations by integrating physicochemical, sequence-context, and structure-informed features. All three modalities are encoded independently and subsequently fused.

Physicochemical encoding

Residue i is encoded using a 20-dim physicochemical descriptor vector $\mathbf{S}_i^{phys} \in \mathbb{R}^{d_{phys}}$ derived from AAindex³¹ (selected properties, such as hydrophobicity, charge, polarity, molecular volume, and side-chain flexibility; z-score normalized across amino acids) (Supplementary Table 1). For a sequence of length L , physicochemical features are assembled into a matrix

$$\mathbf{S}^{phys} = [\mathbf{S}_1^{phys}, \dots, \mathbf{S}_L^{phys}]^T \in \mathbb{R}^{L \times d_{phys}}, d_{phys} = 20. \quad (1)$$

To align this modality with the model's latent space, we apply a fully connected layer to map the features to d dimensions:

$$\mathbf{X}^{phys} = \text{Linear}(\mathbf{S}^{phys}) \in \mathbb{R}^{L \times d}, d = 256. \quad (2)$$

Sequence context encoding

To capture evolutionary and contextual information beyond local amino acid properties, we employ a pretrained protein language model, ESM2 (650 million parameters)³². For an input sequence, residue-level semantic embeddings are extracted from the final transformer layer of ESM2 followed by a fully connected layer:

$$\mathbf{S}^{sem} = \text{ESM2}(\mathbf{S}), \mathbf{S}^{sem} \in \mathbb{R}^{L \times d_{sem}}, d_{sem} = 1280. \quad (3)$$

$$\mathbf{X}^{sem} = \text{Linear}(\mathbf{S}^{sem}) \in \mathbb{R}^{L \times d}, d = 256. \quad (4)$$

Structure geometric encoding

Protein structural information is derived from 3D structures predicted by ESMFold. For residue i , we extract a structural descriptor vector $\mathbf{X}_i^{struct} \in \mathbb{R}^{d_{struct}}$, $d_{struct} = 17$, which includes backbone torsion angles (encoded as sine and cosine pairs), secondary structure states, relative solvent accessibility, residue contact counts, and predicted confidence

scores (pLDDT). In addition, the Cartesian coordinates of the C α atom are retained: $\mathbf{r}_i \in \mathbb{R}^3$.

To explicitly model spatial relationships, residues are represented as nodes in a geometric graph, and a stacked SE(3)-equivariant graph neural network (EGNN) is applied. For EGNN layer l , node features and coordinates are updated as:

$$\mathbf{m}_{ij} = \phi_m(\mathbf{h}_i^{(l)}, \mathbf{h}_j^{(l)}, \|\mathbf{r}_i^{(l)} - \mathbf{r}_j^{(l)}\|^2), \quad (5)$$

$$\mathbf{r}_i^{(l+1)} = \mathbf{r}_i^{(l)} + \sum_{j \neq i} (\mathbf{r}_i^{(l)} - \mathbf{r}_j^{(l)}) \phi_r(\mathbf{m}_{ij}), \quad (6)$$

$$\mathbf{m}_i = \sum_{j \neq i} \mathbf{m}_{ij}, \quad (7)$$

$$\mathbf{h}_i^{(l+1)} = \phi_h(\mathbf{h}_i^{(l)}, \mathbf{m}_i), \quad (8)$$

where ϕ_m , ϕ_r , and ϕ_h are learnable multilayer perceptrons. Node features are initialized as $\mathbf{h}_i^{(0)} = \mathbf{X}_i^{struct}$ and $\mathbf{r}_i^{(0)} = \mathbf{r}_i$. The coordinate update formulation guarantees SE(3)-equivariance.

Finally, the structure-derived geometric encoding is defined as the node embeddings from the final layer N followed by a fully connected layer:

$$\mathbf{S}^{struct} = \mathbf{h}^{(N)} \in \mathbb{R}^{L \times d_{egnn}}, \quad (9)$$

$$\mathbf{X}^{struct} = \text{Linear}(\mathbf{S}^{struct}) \in \mathbb{R}^{L \times d}, d = 256, \quad (10)$$

where d_{egnn} denotes the hidden dimension of the EGNN.

Multimodal feature fusion

Physicochemical and sequence-context features are fused using a gated cross-attention mechanism. Given $\mathbf{X}^{phys}$ and $\mathbf{X}^{sem}$, cross-attention is computed as:

$$\mathbf{A} = \text{Attn}(\mathbf{Q} = \mathbf{X}^{phys}, \mathbf{K} = \mathbf{X}^{sem}, \mathbf{V} = \mathbf{X}^{sem}), \quad (11)$$

where

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V}. \quad (12)$$

A learnable gating function controls modality contribution:

$$\omega_1 = \sigma\left(\text{Linear}\left(\left[\mathbf{X}^{phys}; \mathbf{A}\right]\right)\right), \quad (13)$$

$$\mathbf{E}^{seq} = \text{LayerNorm}\left(\omega_1 \cdot \mathbf{X}^{phys} + (1 - \omega_1) \cdot \mathbf{A}\right), \quad (14)$$

Structure-aware embeddings are further integrated with sequence-derived features. The final residue representation is computed as:

$$\omega_2 = \sigma\left(\text{Linear}\left(\left[\mathbf{E}^{seq}; \text{Attn}(\mathbf{E}^{seq}, \mathbf{X}^{struct}, \mathbf{X}^{struct})\right]\right)\right), \quad (15)$$

$$\mathbf{E}^{fused} = \omega_2 \cdot \mathbf{E}^{seq} + (1 - \omega_2) \cdot \text{Attn}(\mathbf{E}^{seq}, \mathbf{X}^{struct}, \mathbf{X}^{struct}). \quad (16)$$

where $\omega_k \in \mathbb{R}$, $k \in \{1, 2\}$ is a scalar gating coefficient obtained via a sigmoid activation function $\sigma(\cdot)$.

Interaction modeling: bilinear attention networks (BAN)

To model the fine-grained contact interface and capture the intricate dependencies between the residues of two protein chains (e.g., Peptide and HLA, or TCR and pHLA), we implement Bilinear Attention

Networks. Unlike standard dot-product attention which often reduces information into a single dimension, BAN models pairwise residue interactions via a bilinear compatibility function and aggregates interaction-aware features without explicitly constructing a full outer-product tensor.

We utilize learnable weight matrices $\mathbf{U} \in \mathbb{R}^{D \times D'}$ and $\mathbf{V} \in \mathbb{R}^{D \times D'}$ to project two representations $\mathbf{E} \in \mathbb{R}^{L_e \times D}$ and $\mathbf{T} \in \mathbb{R}^{L_t \times D}$ into a joint space of dimension $D' = 256$. The transformed features for the i -th residue of $\mathbf{E}$ and the j -th residue of $\mathbf{T}$ are:

$$\tilde{\mathbf{e}}_i = \mathbf{U}^\top \mathbf{E}_i, \quad \tilde{\mathbf{t}}_j = \mathbf{V}^\top \mathbf{T}_j, \quad (17)$$

where $\tilde{\mathbf{e}}_i, \tilde{\mathbf{t}}_j \in \mathbb{R}^{D'}$.

The interaction score $\mathbf{S}_{i,j}$ for each residue pair is computed using a learnable parameter $\mathbf{q} \in \mathbb{R}^{D'}$ to measure bilinear compatibility as:

$$\mathbf{S}_{i,j} = \tilde{\mathbf{e}}_i^\top \cdot \text{diag}(\mathbf{q}) \cdot \tilde{\mathbf{t}}_j, \quad (18)$$

where $\mathbf{S} \in \mathbb{R}^{L_e \times L_t}$.

The attention map $\mathbf{I}_{i,j}$ is normalized via an explicit 2D Softmax function:

$$\mathbf{I}_{i,j} = \frac{\exp(\mathbf{S}_{i,j})}{\sum_{a=1}^{L_e} \sum_{b=1}^{L_t} \exp(\mathbf{S}_{a,b})}, \quad (19)$$

where $\mathbf{I} \in \mathbb{R}^{L_e \times L_t}$.

The pooled interaction vector $\mathbf{z}$ is derived through the Hadamard product ($\odot$) across all residue pairs:

$$\mathbf{z} = \text{BatchNorm}\left(\sum_{i=1}^{L_e} \sum_{j=1}^{L_t} \mathbf{I}_{i,j} \cdot (\tilde{\mathbf{e}}_i \odot \tilde{\mathbf{t}}_j)\right), \quad (20)$$

where $\mathbf{z} \in \mathbb{R}^{D'}$.

Peptide-HLA predictor model

The peptide-HLA predictor integrates SSFE, Transformer encoders, cross-attention, and BAN.

Let $\mathbf{S}^{pep}$ and $\mathbf{S}^{hla}$ denote the amino acid sequence of peptide and HLA. The model accepts variable-length peptide inputs through residue-level sequence encoding and attention-based interaction modeling, without requiring peptides to be fixed to 9-mers; peptide length is handled implicitly through padding and masking during batch processing. Peptide and HLA residue features are first encoded independently by SSFE module:

$$\mathbf{E}^{fused:pep} = \text{SSFE}(\mathbf{S}^{pep}), \quad \mathbf{E}^{fused:hla} = \text{SSFE}(\mathbf{S}^{hla}), \quad (21)$$

and Transformer blocks:

$$\mathbf{E}^{pep} = \text{Transformer}(\mathbf{E}^{fused:pep}), \quad \mathbf{E}^{hla} = \text{Transformer}(\mathbf{E}^{fused:hla}). \quad (22)$$

Bidirectional cross-attention is then applied:

$$\mathbf{P} = \text{LayerNorm}\left(\mathbf{E}^{pep} + \text{Attn}\left(\mathbf{E}^{pep}, \mathbf{E}^{hla}, \mathbf{E}^{hla}\right)\right), \quad (23)$$

$$\mathbf{H} = \text{LayerNorm}\left(\mathbf{E}^{hla} + \text{Attn}\left(\mathbf{E}^{hla}, \mathbf{E}^{pep}, \mathbf{E}^{pep}\right)\right). \quad (24)$$

BAN is applied between peptide residues and HLA pocket residues predefined as 34 canonical binding-groove positions¹¹, producing an

interaction vector,

$$\mathbf{z}_{pHLA} = \text{BAN}(\mathbf{P}, \mathbf{H}), \quad (25)$$

which is passed to a multilayer perceptron with a sigmoid activation to predict peptide-HLA binding score. Importantly, while the full pHLA model natively supports multimodal structural fusion via the SSFE module, the peptide's structural branch can be optionally deactivated during training and inference. This flexible design maximizes computational throughput during ultra-large-scale virtual screening (e.g., proteome-wide screening) where sequence and physicochemical features are highly sufficient for presentation modeling, while the complete tri-molecular structural architecture is fully engaged during the downstream TCR recognition task.

TCR-peptide-HLA predictor model

The TCR-pHLA predictor model predicts the specific recognition of a pHLA complex by a TCR α/β heterodimer. This model utilizes a multi-stage training pipeline and a sophisticated interaction hierarchy to handle the increased complexity of the quaternary complex.

Feature aggregator: pre-training and transfer learning

To stabilize and guide the learning of TCR-pHLA recognition, the pHLA predictor is first pre-trained and subsequently reused as a feature aggregator. This transfer learning strategy ensures that the model leverages established knowledge of pHLA complex stability before modeling the TCR interaction.

Let $\mathbf{H}^{pre}$ and $\mathbf{P}^{pre}$ denote the pre-trained embeddings (for HLA and peptide, respectively) extracted from the frozen pHLA predictor, and let $\mathbf{H}^{task}$ and $\mathbf{P}^{task}$ denote the task-specific embeddings generated within the TCR-pHLA pipeline. These representations are integrated through gated fusion:

$$\omega_{hla} = \sigma\left(\text{Linear}\left(\left[\mathbf{H}^{task}; \text{Attn}\left(\mathbf{H}^{task}, \mathbf{H}^{pre}, \mathbf{H}^{pre}\right)\right]\right)\right), \quad (26)$$

$$\omega_{pep} = \sigma\left(\text{Linear}\left(\left[\mathbf{P}^{task}; \text{Attn}\left(\mathbf{P}^{task}, \mathbf{P}^{pre}, \mathbf{P}^{pre}\right)\right]\right)\right), \quad (27)$$

$$\mathbf{H}^{agg} = \text{LayerNorm}(\omega_{hla} \cdot \mathbf{H}^{task} + (1 - \omega_{hla}) \cdot \text{Attn}(\mathbf{H}^{task}, \mathbf{H}^{pre}, \mathbf{H}^{pre})), \quad (28)$$

$$\mathbf{P}^{agg} = \text{LayerNorm}(\omega_{pep} \cdot \mathbf{P}^{task} + (1 - \omega_{pep}) \cdot \text{Attn}(\mathbf{P}^{task}, \mathbf{P}^{pre}, \mathbf{P}^{pre})), \quad (29)$$

The pretrained representations $\mathbf{H}^{pre}$ and $\mathbf{P}^{pre}$ provide a presentation-aware representation of the pHLA complex, including peptide-HLA compatibility and HLA-restricted context. In contrast, $\mathbf{H}^{task}$ and $\mathbf{P}^{task}$ are optimized directly for TCR recognition. This design does not assume that TCR binding is driven by the same residues or features that determine peptide-HLA presentation. Instead, the pre-trained branch provides a stable pHLA-context prior, while the task-specific branch learns features associated with TCR engagement. These two representations are combined through an input-dependent gating mechanism, which controls their relative contribution for each pHLA input and allows the model to balance presentation-aware context with TCR-recognition-specific information.

Interaction hierarchy: SSFE + transformer + cross attention + BAN

Following feature aggregation, the TCR-pHLA interaction is modeled through a hierarchical sequence of operations:

- Let $\mathbf{S}^{tcr\alpha}$ and $\mathbf{S}^{tcr\beta}$ denote the amino acid sequences of TCR α and TCR β chains. The residue features are encoded independently by

SSFE module and Transformer blocks:

$$\mathbf{E}^{tcr\alpha} = \text{Transformer}(\text{SSFE}(\mathbf{S}^{tcr\alpha})), \mathbf{E}^{tcr\beta} = \text{Transformer}(\text{SSFE}(\mathbf{S}^{tcr\beta})) \quad (30)$$

- CDR3 Segment Extraction:** The model identifies and extracts the hypervariable CDR3 α and CDR3 β segments. For each chain, let $i_{start}^\alpha, i_{end}^\alpha$ and $i_{start}^\beta, i_{end}^\beta$ be the anchor residue indices. The segment $\mathbf{E}^{cdr3a}$ and $\mathbf{E}^{cdr3b}$ is extracted as:

$$\mathbf{E}^{cdr3a} = \mathbf{E}^{tcr\alpha}[i_{start}^\alpha : i_{end}^\alpha], \mathbf{E}^{cdr3b} = \mathbf{E}^{tcr\beta}[i_{start}^\beta : i_{end}^\beta]. \quad (31)$$

- Multichannel Cross-Attention:** Mutual information is exchanged between the TCR loops and the pHLA aggregator features to refine the docking orientation:

$$\mathbf{T}^{cdr3a} \leftarrow \text{LayerNorm}(\mathbf{E}^{cdr3a} + \text{Attn}(\mathbf{E}^{cdr3a}, \mathbf{P}^{agg}, \mathbf{P}^{agg})), \quad (32)$$

$$\mathbf{T}^{cdr3b} \leftarrow \text{LayerNorm}(\mathbf{E}^{cdr3b} + \text{Attn}(\mathbf{E}^{cdr3b}, \mathbf{P}^{agg}, \mathbf{P}^{agg})), \quad (33)$$

$$\mathbf{T}^{tcr\alpha} \leftarrow \text{LayerNorm}(\mathbf{E}^{tcr\alpha} + \text{Attn}(\mathbf{E}^{tcr\alpha}, \mathbf{H}^{agg}, \mathbf{H}^{agg})), \quad (34)$$

$$\mathbf{T}^{tcr\beta} \leftarrow \text{LayerNorm}(\mathbf{E}^{tcr\beta} + \text{Attn}(\mathbf{E}^{tcr\beta}, \mathbf{H}^{agg}, \mathbf{H}^{agg})), \quad (35)$$

- Bilinear Interaction Modeling:** Using the BAN architecture described in Section 3.3, the model computes four distinct interaction vectors:

$$\mathbf{z}_{cdr3a-pep} = \text{BAN}(\mathbf{T}^{cdr3a}, \mathbf{P}^{agg}), \mathbf{z}_{cdr3b-pep} = \text{BAN}(\mathbf{T}^{cdr3b}, \mathbf{P}^{agg}), \quad (36)$$

$$\mathbf{z}_{tcr\alpha-hla} = \text{BAN}(\mathbf{T}^{tcr\alpha}, \mathbf{H}^{agg}), \mathbf{z}_{tcr\beta-hla} = \text{BAN}(\mathbf{T}^{tcr\beta}, \mathbf{H}^{agg}). \quad (37)$$

- Prediction Head:** These vectors are concatenated into a global interaction representation $\mathbf{z}_{total}$ and passed through a multi-layer perceptron (MLP) with a sigmoid activation to output the final TCR-pHLA binding score.

Negative sampling

To improve discrimination and mitigate shortcut learning under limited experimentally validated negatives, we used two complementary negative-sampling strategies: dynamic shuffling-based negatives and mutation-aware hard negatives.

Dynamic negative sampling

Because comprehensive negative binding data are scarce for TCR-pHLA interaction prediction, we generated negative pairs for training. Common strategies include: (i) fixing pHLA and randomly shuffling TCRs to form mismatched pairs, while excluding any known positive pairs; and (ii) pairing pHLA with background TCRs from healthy donors. We adopted the shuffling-based strategy to avoid potential distribution mismatches between positive and negative TCR repertoires that may arise when using healthy-donor backgrounds³⁹. Concretely, we sampled negatives at a 3:1 negative-to-positive ratio. Rather than fixing a single set of negatives before training, we regenerated shuffled negatives dynamically during training (e.g., each epoch),

which expands the effective negative space while keeping the positive set unchanged and reduces reliance on a limited, static negative pool.

Mutation-aware hard negative sampling

In the cancer mutation analyses, we additionally introduced a mutation-aware strategy to construct hard negatives that more closely reflect the biological challenge of distinguishing mutants from closely related self-contexts. Specifically, we generated decoy pairs via small sequence perturbations:

- **Peptide mutation:** for a positive peptide, we randomly substitute 1–2 residues to create minimally perturbed decoys that are expected to reduce peptide–HLA and/or TCR recognition compatibility.
- **TCR CDR3 mutation:** we randomly mutate 1–3 residues within the CDR3 loop(s) (excluding conserved anchor positions) to produce near-neighbor decoys likely to alter specificity.

We note that this strategy can introduce label noise, as a subset of perturbed sequences may still retain measurable binding. Nevertheless, under this construction, the experimentally supported positives are expected to exhibit higher binding propensity on average than the mutated decoys, making these hard negatives a useful approximation for training and evaluation in the absence of exhaustive negative measurements.

Training objective

Models are trained using focal loss to address class imbalance. For label $y \in \{0, 1\}$ and predicted probability p , the loss is defined as:

$$L = -\alpha y(1-p)^{\gamma} \log(p) - (1-\alpha)(1-y)p^{\gamma} \log(1-p), \quad (38)$$

where α balances positive and negative classes and γ emphasizes hard examples. This strategy forces the model to distinguish true molecular recognition from highly similar decoy interactions, substantially improving specificity.

Training details

- **Optimizer:** AdamW with a learning rate of 1×10^{-4} and weight decay.
- **Regularization:** Dropout of 0.1 and Gradient Clipping at a max norm of 2.0.
- **Cross-Validation:** 5-fold cross-validation with an EarlyStopping patience of 5 epochs for Peptide–HLA predictor, and 10-fold cross-validation with an EarlyStopping patience of 10 epochs for TCR–pHLA predictor, both based on validation AUROC. The maximum epoch was set to 100.
- **Batch size:** The batch size was set to 256.
- **Hardware:** All models were trained on NVIDIA A6000 GPUs using the PyTorch framework.

Evaluation metrics

We evaluated model performance using AUROC, AUPRC, recall, true discovery rate (TDR), and AUC0.1. The AUROC (area under the receiver operating characteristic curve) was used to measure the ability of a model to rank positive interactions above negative interactions across all possible classification thresholds. Because many immunological prediction tasks are highly imbalanced, we also reported AUPRC (the area under the precision–recall curve), which is more sensitive to the recovery of rare positive interactions among a large number of negatives.

For candidate prioritization analyses, recall was defined as the fraction of experimentally validated positives recovered within the top-ranked predictions. TDR was defined as the fraction of top-ranked candidates that were experimentally validated positives, providing a

practical measure of enrichment among candidates selected for downstream validation.

AUC0.1 was used for the IMMREP23 benchmark following the original challenge protocol. It represents the partial area under the ROC curve calculated only up to a false positive rate of 0.1. This metric focuses on the high-specificity region of the ROC curve and is therefore useful for evaluating models in settings where false positives are costly and only a small number of predicted TCR–epitope interactions can be experimentally tested.

Modality-level SHAP analysis

To demystify the multi-modal architecture and quantitatively assess the global importance of distinct data modalities and biological components, we implemented a customized, exact SHAP (SHapley Additive exPlanations) framework. Standard SHAP estimation packages (e.g., KernelSHAP) are generally designed for high-dimensional feature spaces and often rely on approximations. Given our objective to evaluate a constrained number of macro-level features (i.e., data modalities or biological components, denoted as M), we bypassed approximation algorithms and computed the exact Shapley values to ensure maximum mathematical rigor.

Specifically, for each sample in the test set, we evaluated the model's predictive output across all 2^M possible combinations (coalitions) of input modalities. The masked inputs were substituted with expected values drawn from a background dataset distribution.

The exact Modality SHAP value ϕ_i for the i -th modality was then calculated as the weighted average of its marginal contributions to all possible subsets S that do not contain i :

$$\phi_i = \sum_{S \subseteq M \setminus \{i\}} \frac{|S|!(|M| - |S| - 1)!}{|M|!} (\nu(S \cup \{i\}) - \nu(S)), \quad (39)$$

where $\nu(S)$ represents the model's prediction given the subset of active modalities S . This exact ablation-based approach provides a highly robust quantification of each module's contribution to the inferred TCR–pHLA interactions.

Experimental details

Plasmid construction. The AS8.4 TCR sequences TRBV901–TRBJ2–301 and TRAV2101–TRAJ2801 were obtained from the international ImMunoGeneTics information system (IMGT/GENE-DB, <https://www.imgt.org/>). CDR3 regions correspond to those reported by Yang et al. (β : CASSVGTYSTDTQYF; α : CAVNSPGSGAGSYQLTF). Gene fragments were codon-optimized for Homo sapiens, joined with a P2A linker, and assembled into a pLX307 vector. Correct assembly was confirmed by Sanger sequencing (CCIB DNA Core, Massachusetts General Hospital, Boston).

Cell lines. Jurkat E6.1 (ATCC; TIB-152), HEK293T (ATCC; CRL-3216), and Plat-A (Cell Biolabs, San Diego, CA, USA; RV-102) cells were cultured at 37 °C and 5 % CO₂. Jurkat and Plat-A cells were maintained in RPMI-1640 (Thermo Fisher; 61870036); HEK293T and Plat-A cells were maintained in DMEM (Thermo Fisher; 10569010). Media were supplemented with 10 % (v/v) heat-inactivated FBS (Thermo Fisher; 26140079) and 100 $\mu\text{g ml}^{-1}$ penicillin–streptomycin (Thermo Fisher; 15140-122). Cell lines were screened mycoplasma-free every 3 months with the Mycoplasma Detection Kit (InvivoGen; rep-mys-50).

CRISPR–Cas9 knockout of endogenous TCR. To eliminate endogenous TCR expression in Jurkat cells, both the TCR α constant region (TRAC) and the TCR β constant region (TRBC1) were targeted using synthetic guide RNAs (sgRNAs) complexed with Cas9–GFP protein (IDT; 10008100). Ribonucleoprotein complexes were assembled at a 1:2 molar ratio of Cas9–GFP (IDT; 10008100) to sgRNA (IDT). The following sgRNA sequences were used: for TRAC, 5'-

TGGATTAGAGTCTCTCAGC-3' and 5'-ACAAACTGTGCTAGACATG-3'; for TRBCL, 5'-CGTAGAAGTGGACTTGACAG-3' and 5'-CCCAC-CAGCTCAGCTCCACG-3'. Jurkat cells were electroporated with ribonucleoprotein complexes using a 4D-Nucleofector™ X Unit (Lonza) and the SE Cell Line 4D-Nucleofector™ X Kit S (Lonza; V4XC-1032). Program CL-120 was used, and cells were transferred immediately to 2 mL pre-warmed RPMI + 10 % FBS. After 5 days, cells were stained with PE-conjugated anti-human TCRαβ antibody (clone IP26; BioLegend, catalog no. 306708) at a 1:100 dilution in a total staining volume of 100 μL for 30 min on ice. TCRαβ-negative cells were sorted using a Bigfoot™ Spectral Cell Sorter (Thermo Fisher Scientific), and the post-sort purity was ≥95.0%.

Lentiviral vector production and transduction. HEK293T cells were co-transfected with the plasmids psPAX2 (Addgene #12260) and pMD2.G (Addgene #12259) at 4:4:1 ratio using TransIT®-LT1 (Mirus Bio; MIR 2300). Supernatants were harvested after 72 h, filtered, and stored at -80 °C. Viral concentrations were determined with Lenti-X™ GoS-tix™ Plus (Takara Bio, 631280). TCR knockout Jurkat cells were transduced with the virus by spinfection (1000 × g, 90 min, 32 °C) in the presence of 8 μg/mL polybrene (Sigma-Aldrich; TR-1003-G). After 48 h, 2 μg/mL puromycin was applied for 5 d. Expression of the AS8.4 TCR was verified by staining with PE-conjugated anti-human TCRαβ antibody (clone IP26; BioLegend, catalog no. 306708) and APC-conjugated anti-human CD3 antibody (clone UCHT1; BioLegend, catalog no. 300411). Both antibodies were used at a 1:100 dilution in a total staining volume of 100 μL, and cells were incubated for 30 min on ice.

Retroviral delivery of CD8αβ and NFAT-ZsGreen reporter. The CD8α-CD8β-8×NFAT-ZsGreen construct (Addgene #153417) was packaged in Plat-A cells. TransIT-LT1 (Mirus Bio; MIR 2300) was used for transfection. Supernatants (48 h, 72 h) were filtered and concentrated 10-fold with the Lenti-X Concentrator (Takara Bio; 631231). Jurkat-AS8.4 cells were spin-infected as described above and cultured for 72 h. Cells were then stained with PerCP-conjugated anti-human CD8α antibody (clone RPA-T8; BioLegend, catalog no. 301030) at a 1:100 dilution in a total staining volume of 100 μL for 30 min on ice. CD8α-positive cells were sorted, and the post-sort purity was ≥95.0%.

Generation of HEK293T-B2M-HLA-B*27:05 antigen-presenting cells. Lentiviral construct encoding human β₂-microglobulin (B2M) and HLA-B*27:05 was packaged as above. HEK293T cells were transduced by spinfection (1000 × g, 90 min, 32 °C) in the presence of 8 μg/mL polybrene, selected with puromycin (2 μg/mL, 5 d), and cell-surface expression was assessed by staining with FITC-conjugated mouse anti-human HLA-B27 antibody (clone HLA-ABC-m3; Bio-Rad, catalog no. MCA116F) at a 1:100 dilution in a total staining volume of 100 μL for 30 min on ice.

Peptide stimulation assay. HEK293T-HLA-B*27 cells (1 × 10⁵ cells per well) were seeded in 96-well U-bottom plates in 100 μL DMEM complete medium. After 24 h, synthetic peptides (GenScript; HPLC purity ≥95.0%) were dissolved in water and added at a final concentration of 100 μM. Peptides were pre-incubated with HEK293T-HLA-B*27 cells for 60 min at 37 °C and 5 % CO₂. Then 1 × 10⁵ Jurkat-AS8.4-NFAT-ZsGreen cells in 100 μL RPMI + 10 % FBS were added. Co-cultures proceeded 24 h (37 °C, 5 % CO₂).

Cells were harvested and stained with APC-conjugated anti-human CD69 antibody (clone FN50; BioLegend, catalog no. 985206) at a 1:100 dilution in a total volume of 100 μL PBS supplemented with 2% FBS for 30 min on ice (Supplementary Table 7). Following staining, cells were washed and resuspended in 300 μL PBS supplemented with 2% FBS. Data were acquired using a Cytex® Aurora spectral flow cytometer (Cytex Biosciences, Fremont, CA, USA) with SpectroFlo v3.0.3 and analyzed using FlowJo™ v10.10.0 (BD Biosciences, San Jose, CA,

USA). Jurkat cells were identified by forward- and side-scatter characteristics, followed by singlet gating. T-cell activation was quantified as the percentage of CD69-positive and NFAT-ZsGreen-positive cells among live singlet Jurkat cells. The peptide stimulation assay was performed in two independent experiments.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The final model weights, training data and AS-related microbial peptides analyzed in this study are publicly available via Zenodo at (<https://doi.org/10.5281/zenodo.18002170>). The benchmarking datasets analyzed in this study are publicly available through their original publications, including the Melanoma dataset² (<https://doi.org/10.1016/j.cell.2025.06.019>), Albert et al.¹² (<https://doi.org/10.1038/s42256-023-00694-6>), Chu et al.¹³ (<https://doi.org/10.1038/s42256-022-00459-7>), IMMREP22¹⁵ (<https://doi.org/10.1016/j.immuno.2023.100024>), IMMREP23¹⁶ (<https://www.kaggle.com/competitions/tcr-specificity-prediction-challenge>), Lu et al.¹⁷ (<https://doi.org/10.1038/s41592-025-02910-0>), Que et al.²² (<https://doi.org/10.1038/s41467-025-60461-6>), TESLA³⁶ (<https://doi.org/10.1016/j.cell.2020.09.015>), CEDAR³⁷ (<https://cedar.iedb.org>), Minervina et al.³⁸ (<https://doi.org/10.1038/s41590-022-01184-4>), Zhang et al.²³ (<https://doi.org/10.1038/s42256-024-00913-8>), TP53 dataset⁴³ (<https://doi.org/10.1158/2326-6066.CIR-22-0040>). All other data are available in the article and its Supplementary files or from the corresponding author upon request. Source data are provided with this paper.

Code availability

The open-source StriMap codebase is maintained at (<https://github.com/uhrerlab/strimap-tools>) and Zenodo at <https://doi.org/10.5281/zenodo.21925295> (ref.⁵⁶). A publicly accessible web server implementing the StriMap prediction platform is available at (www.strimap.com), providing batch prediction, analysis, visualization and model fine tuning for peptide-HLA binding and TCR-pHLA recognition.

References

- Punt, J., Stranford, S., Jones, P., & Owen, J. *Kuby Immunology*. (Macmillan Higher Education, 2018).
- Blass, E. et al. A multi-adjuvant personal neoantigen vaccine generates potent immunity in melanoma. *Cell* **188**, 5125–5141.e27 (2025).
- Klebanoff, C. A., Chandran, S. S., Baker, B. M., Quezada, S. A. & Ribas, A. T cell receptor therapeutics: immunological targeting of the intracellular cancer proteome. *Nat. Rev. Drug Discov.* **22**, 996–1017 (2023).
- Theofilopoulos, A. N., Kono, D. H. & Baccala, R. The multiple pathways to autoimmunity. *Nat. Immunol.* **18**, 716–724 (2017).
- Buckner, J. H. Antigen-specific immunotherapies for autoimmune disease. *Nat. Rev. Rheumatol.* **21**, 88–97 (2025).
- Zhang, S.-Q. et al. High-throughput determination of the antigen specificities of T cell receptors in single cells. *Nat. Biotechnol.* **36**, 1156–1159 (2018).
- Kula, T. et al. T-scan: a genome-wide method for the systematic discovery of T cell epitopes. *Cell* **178**, 1016–1028 (2019).
- Joglekar, A. V. & Li, G. T cell antigen discovery. *Nat. Methods* **18**, 873–880 (2021).
- Hudson, D., Fernandes, R. A., Basham, M., Ogg, G. & Koohy, H. Can we predict T cell specificity with digital biology and machine learning? *Nat. Rev. Immunol.* **23**, 511–521 (2023).
- Nilsson, J. B., Greenbaum, J., Peters, B. & Nielsen, M. NetMHCpan-4.2: improved prediction of CD8+ epitopes by use of transfer learning and structural features. *Front. Immunol.* **16**, 1616113 (2025).

11. Reynisson, B., Alvarez, B., Paul, S., Peters, B. & Nielsen, M. NetMHCpan-4.1 and NetMHCIIpan-4.0: improved predictions of MHC antigen presentation by concurrent motif deconvolution and integration of MS MHC eluted ligand data. *Nucleic Acids Res.* **48**, W449–W454 (2020).
12. Albert, B. A. et al. Deep neural networks predict class I major histocompatibility complex epitope presentation and transfer learn neoepitope immunogenicity. *Nat. Mach. Intell.* **5**, 861–872 (2023).
13. Chu, Y. et al. A transformer-based model to predict peptide–HLA class I binding and optimize mutated peptides for vaccine design. *Nat. Mach. Intell.* **4**, 300–311 (2022).
14. Stražar, M. et al. HLA-II immunopeptidome profiling and deep learning reveal features of antigenicity to inform antigen discovery. *Immunity* **56**, 1681–1698.e13 (2023).
15. Meysman, P. et al. Benchmarking solutions to the T-cell receptor epitope prediction problem: IMMREP22 workshop report. *Immunoinformatics (Amst.)* **9**, 100024 (2023).
16. Nielsen, M. et al. Lessons learned from the IMMREP23 TCR-epitope prediction challenge. *Immunoinformatics (Amst.)* **16**, 100045 (2024).
17. Lu, Y. et al. Assessment of computational methods in predicting TCR-epitope binding recognition. *Nat. Methods* **23**, 248–259 (2026).
18. Montemurro, A. et al. NetTCR-2.0 enables accurate prediction of TCR-peptide binding by using paired TCR α and β sequence data. *Commun. Biol.* **4**, 1060 (2021).
19. Jensen, M. F. & Nielsen, M. NetTCR 2.2 - Improved TCR specificity predictions by combining pan- and peptide-specific training strategies, loss-scaling and integration of sequence similarity. *eLife* **12**, RP93934 (2024).
20. Lu, T. et al. Deep learning-based prediction of the T cell receptor-antigen binding specificity. *Nat. Mach. Intell.* **3**, 864–875 (2021).
21. Gao, Y. et al. Pan-Peptide Meta Learning for T-cell receptor–antigen binding recognition. *Nat. Mach. Intell.* **5**, 236–249 (2023).
22. Que, J. et al. Identifying T cell antigen at the atomic level with graph convolutional network. *Nat. Commun.* **16**, 5171 (2025).
23. Zhang, Y. et al. Epitope-anchored contrastive transfer learning for paired CD8⁺ T cell receptor–antigen recognition. *Nat. Mach. Intell.* **6**, 1344–1358 (2024).
24. Croce, G. et al. Deep learning predictions of TCR-epitope interactions reveal epitope-specific chains in dual alpha T cells. *Nat. Commun.* **15**, 3211 (2024).
25. Fast, E., Dhar, M. & Chen, B. TAPIR: a T-cell receptor language model for predicting rare and novel targets. *bioRxiv* <https://doi.org/10.1101/2023.09.12.557285> (2023).
26. Li, X., Sun, C., Huang, W., Wang, Y. & Ma, B. SageTCR: a structure-based model integrating residue- and atom-level representations for enhanced TCR-pMHC binding prediction. *Brief. Bioinform.* **26**, bbaf496 (2025).
27. Deleuran, S. N. & Nielsen, M. NetTCR-struc, a structure driven approach for prediction of TCR-pMHC interactions. *Front. Immunol.* **16**, 1616328 (2025).
28. Ge, J. et al. TRAP: a contrastive learning-enhanced framework for robust TCR-pMHC binding prediction with improved generalizability. *Chem. Sci.* **16**, 9881–9894 (2025).
29. Slone, J. K., Conev, A., Rigo, M. M., Reuben, A. & Kaviraki, L. E. TCR-pMHC binding specificity prediction from structure using graph neural networks. *IEEE Trans. Comput. Biol. Bioinform.* **22**, 171–179 (2025).
30. Slone, J. K. et al. STAG-LLM: predicting TCR-pHLA binding with protein language models and computationally generated 3D structures. *Comput. Struct. Biotechnol. J.* **27**, 3885–3896 (2025).
31. Kawashima, S. & Kanehisa, M. AAindex: amino acid index database. *Nucleic Acids Res.* **28**, 374 (2000).
32. Lin, Z. et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* **379**, 1123–1130 (2023).
33. Elnaggar, A. et al. ProtTrans: toward understanding the language of life through self-supervised learning. *IEEE Trans. Pattern Anal. Mach. Intell.* **44**, 7112–7127 (2022).
34. Garcia Satorras, V., Hoogeboom, E. & Welling, M. E(n) equivariant graph neural networks. *Proc. 38th Int. Conf. Mach. Learn. PMLR* **139**, 9323–9332 (2021).
35. Kim, J.-H., Jun, J. & Zhang, B.-T. *Bilinear Attention Networks*. in *Proceedings of the 32nd International Conference on Neural Information Processing Systems* 1571–1581 (Curran Associates Inc., Red Hook, NY, USA, 2018).
36. Wells, D. K. et al. Key parameters of tumour epitope immunogenicity revealed through a consortium approach improve neoantigen prediction. *Cell* **183**, 818–834.e13 (2020).
37. Koşaloğlu-Yalçın, Z. et al. The Cancer Epitope Database and Analysis Resource (CEDAR). *Nucleic Acids Res.* **51**, D845–D852 (2023).
38. Minervina, A. A. et al. SARS-CoV-2 antigen exposure history shapes phenotypes and specificity of memory CD8⁺ T cells. *Nat. Immunol.* **23**, 781–790 (2022).
39. Dens, C., Laukens, K., Bittremieux, W. & Meysman, P. The pitfalls of negative data bias for the T-cell epitope specificity challenge. *Nat. Mach. Intell.* **5**, 1060–1062 (2023).
40. Vita, R. et al. The immune epitope database (IEDB): 2024 update. *Nucleic Acids Res.* **53**, D436–D443 (2025).
41. Shugay, M. et al. VDJdb: a curated database of T-cell receptor sequences with known antigen specificity. *Nucleic Acids Res.* **46**, D419–D427 (2018).
42. Povlsen, H. R. et al. Improved T cell receptor antigen pairing through data-driven filtering of sequencing information from single cells. *Elife* **12**, e81810 (2023).
43. Kim, S. P. et al. Adoptive cellular therapy with autologous tumor-infiltrating lymphocytes and T-cell receptor-engineered T cells targeting common p53 neoantigens in human solid tumors. *Cancer Immunol. Res.* **10**, 932–946 (2022).
44. Yang, X. et al. Autoimmunity-associated T cell receptors recognize HLA-B*27-bound peptides. *Nature* **612**, 771–777 (2022).
45. Thompson, K. N. et al. Alterations in the gut microbiome implicate key taxa and metabolic pathways across inflammatory arthritis phenotypes. *Sci. Transl. Med.* **15**, eabn4722 (2023).
46. Sternes, P. R. et al. Distinctive gut microbiomes of ankylosing spondylitis and inflammatory bowel disease patients suggest differing roles in pathogenesis and correlate with disease activity. *Arthritis Res. Ther.* **24**, 163 (2022).
47. Delgado, L. F. & Andersson, A. F. Evaluating metagenomic assembly approaches for biome-specific gene catalogues. *Microbiome* **10**, 72 (2022).
48. Franzosa, E. A. et al. Gut microbiome structure and metabolic activity in inflammatory bowel disease. *Nat. Microbiol.* **4**, 293–305 (2019).
49. Gevers, D. et al. The treatment-naïve microbiome in new-onset Crohn’s disease. *Cell Host Microbe* **15**, 382–392 (2014).
50. Hall, A. B. et al. A novel Ruminococcus gnavus clade enriched in inflammatory bowel disease patients. *Genome Med* **9**, 103 (2017).
51. Mehta, R. S. et al. Stability of the human faecal microbiome in a cohort of adult men. *Nat. Microbiol.* **3**, 347–355 (2018).
52. Cunningham-Oakes, E. et al. Enhancing infectious intestinal disease diagnosis through metagenomic and metatranscriptomic sequencing of 1000 human diarrhoeal samples. *Genome Med* **17**, 55 (2025).
53. Adams, J. J. et al. T cell receptor signaling is limited by docking geometry to peptide-major histocompatibility complex. *Immunity* **35**, 681–693 (2011).
54. Abramson, J. et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **630**, 493–500 (2024).

55. Wang, N. et al. Deep peptide recognition profiling decodes TCR specificity and enables disease-associated antigen discovery. *Nat. Biotechnol.* (2026).
56. Cao, K. uhlerlab/strimap-tools: strimap-tools first release (Version v1.0.0) [Computer software]. *Zenodo* <https://doi.org/10.5281/zenodo.21925295> (2026).

Acknowledgements

We thank Q. Gong, C. Li, F. Chen, C. Krishna, W. Jin, B. Shao, L. Yuan, X. Zhang, J. Zhang, R. Zhang, and S. Liu for valuable discussions regarding method design. We thank E. Forte, T. Reimels, and H. Kang for helpful discussions and feedback on the manuscript.

Author contributions

R.J.X. and C.U. directed the study. K.C., M.S., E.M.B., and P.N.U.N. conceived the study. K.C. developed the model. K.C., M-M.P. and J.P. performed data analysis. D.B.G. and R.L. designed the AS experiments. R.L. performed the AS experiments. O.A. and D.B.G provided crucial insights. K.C., R.L., M.S., M-M.P., O.A., D.B.G, C.U. and R.J.X. wrote the manuscript.

Funding

K.C. was partially supported by a fellowship from AstraZeneca, the Eric and Wendy Schmidt Center at the Broad Institute of MIT and Harvard, and the National Institutes of Health (DK135492). C.U. was partially supported by NCCIH/NIH (1DP2AT012345), NIDDK/NIH (5RC2DK135492-02), ONR (N00014-24-1-2687), the United States Department of Energy (DE-SC0023187), and MIT J-Clinic for Machine Learning and Health. R.J.X. was supported by the National Institutes of Health (DK043351, AI110495, and DK135492), and The Leona M. and Harry B. Helmsley Charitable Trust.

Competing interests

R.J.X. is a co-founder of Convergence Bio, scientific advisory board member at Nestlé and Magnet BioMedicine, and board director at MoonLake Immunotherapeutics. C.U. serves on the Scientific Advisory Board of Immunai and Relation Therapeutics and has received sponsored research support from AstraZeneca and Janssen Pharmaceuticals. These organizations had no roles in this study. The other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-026-78063-1>.

Correspondence and requests for materials should be addressed to Caroline Uhler or Ramnik J. Xavier.

Peer review information *Nature Communications* thanks Mate Man-czinger and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.